

The Long-Run Dynamics of Relative Price Dispersion^{*}

Fernando Borraz[†] and Leandro Zipitria[‡]

Abstract

This paper studies the long-run increase in same-good retail price dispersion and the changing price positions behind it. Using nearly eighteen years of supermarket prices from Montevideo, Uruguay, we document a persistent increase in dispersion across sellers, despite a partial reversal after 2021–2022. Sellers’ relative positions across products account for 79 percent of the observed increase in across-seller price variance, with a much smaller contribution from differences in sellers’ overall price levels. The widening is already present among retail chains, while average real prices follow a different path. On price spells with observed beginnings, most of the fitted trend in seller-product variance remains when prices are reconstructed at their last nominal adjustment: gross erosion contributes positively but is largely offset by its covariance with reset positions, leaving a limited net contribution on this sample. A multiproduct menu-cost model illustrates a possible monetary implication: economies with identical initial price distributions for every good can respond differently to the same nominal shock when economies of scope in repricing make adjustment depend on how gaps from desired prices are combined within firms. The evidence documents the changing composition of dispersion; the model shows why the organization of price gaps within firms, beyond the amount of dispersion alone, can matter for aggregate price and consumption responses.

JEL Codes: E31, D40, L11, L81.

Keywords: Price Dispersion, Relative Prices, Inflation, Retail Pricing, Price Setting.

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1 Introduction

This paper studies whether the long-run increase in same-good retail price dispersion reflects growing differences in sellers’ overall price levels or in their relative positions across products. Using nearly eighteen years of supermarket prices from Montevideo, Uruguay, we find that the widening primarily occurs in seller-product positions and remains largely present when prices are reconstructed at their last observed nominal adjustment. A multiproduct menu-cost model illustrates a possible monetary implication: identical initial price distributions for every good can coexist with different aggregate responses when economies of scope in repricing make adjustment depend on how gaps are combined within firms.

We extend the component perspective of Kaplan and Menzio (2015) and Kaplan, Menzio, Rudanko, and Trachter (2019) (KMRT) to the long-run evolution of dispersion. Those studies distinguish sellers that are systematically expensive or cheap across products from seller-product relative positions and report store-wide shares of about 10 and 15 percent in U.S. data. In our data, the seller-wide share rises only from 11 to 14 percent, while seller-product positions explain 79 percent of the observed first-to-last increase in across-seller variance and about 90 percent of its fitted trend. Across-seller dispersion itself rises by 2.29 percentage points between the initial and final twelve-month averages, despite a partial reversal after 2021–2022. Category-wide and within-category positions each contribute substantially to the seller-product trend. The long panel therefore adds a dynamic dimension to the composition documented in the U.S. evidence: the widening is driven mainly by divergence in sellers’ relative-price profiles rather than by increasing differences in their overall price levels.

The data cover April 2007–December 2024, 125 consistently identified products, 42 categories, and 61 local markets. Chain affiliation lets us distinguish physical stores from economic sellers. Within-chain dispersion is low and trendless, while widening is already present among chains and is similar in chain–chain and chain–independent comparisons,

connecting the long-run pattern to evidence on centralized and chain-level retail pricing (Borraz and Zipitúa, 2012; Borraz and Saldain, 2017; Klaczko, 2025; Nakamura, Nakamura, and Nakamura, 2011; DellaVigna and Gentzkow, 2019; Hitsch, Hortacısu, and Lin, 2021; Daruich and Kozlowski, 2023). Average real prices follow a different path from dispersion, and the fitted widening is stable to aggregate-inflation controls; category-specific inflation has a small and statistically insignificant contemporaneous coefficient. These comparisons relate to the inflation–dispersion literature (Reinsdorf, 1994; Caglayan, Filiztekin, and Rauh, 2008; Álvarez, Beraja, González-Rozada, and Neumeyer, 2019; Sheremirov, 2020; Sara-Zaror, 2025).

Price histories sharpen the interpretation of the seller-product widening. On observations whose current spell beginning is known, 95.2 percent of the fitted seller-product variance trend remains after reconstructing prices at their most recent nominal adjustment. Gross erosion contributes positively, but its covariance with reset positions offsets most of that contribution; the allocation is conditional on a support with a smaller trend than the full sample. The exercise complements evidence that extreme relative prices can be young (Campbell and Eden, 2014) and differs from reset-price inflation as a measure of changes in desired prices (Bils, Klenow, and Malin, 2012): we reconstruct historical reset values. Adjustment timing also moves along several margins—the upper tail of observed price age lengthens while reset frequency has no significant full-sample trend—rather than through a uniform change in nominal rigidity (Klenow and Kryvtsov, 2008; Nakamura and Steinsson, 2008).

The model asks why the organization of price gaps within multiproduct firms can matter even when product-specific price distributions are identical. Following Bonomo, Carvalho, Kryvtsov, Ribon, and Rigato (2023), firms face a common repricing cost and an additional product-level cost, allowing partial synchronization. When the common cost is large enough that two large gaps jointly activate repricing while one large gap alone does not, economies with common-sign and opposite-sign gaps have different price and consumption responses to the same nominal expenditure shock. Relative shocks

can instead induce universal adjustment and full nominal pass-through. The empirical decomposition measures observed positions rather than desired-price gaps, and the model is an illustration rather than an estimated mapping from those moments to aggregate responses.

Section 2 describes the data. Sections 3–5 establish and decompose the long-run widening and examine price histories. Section 6 presents the multiproduct illustration, and Section 7 concludes.

2 Data

We perform the analysis using a detailed product-level database of daily posted prices compiled by the General Directorate of Commerce (DGC, by its Spanish acronym), a branch of Uruguay’s Ministry of Economy and Finance.¹ Retailers are required to report posted prices under a sworn statement. This reporting obligation does not independently verify every observation or the synchronization of reported changes.

Supermarket sales in Uruguay are concentrated in food, personal-care, and household goods, and supermarket chains account for approximately 70 percent of retail sales (Uruguay XXI, 2018). Chain-level pricing is therefore an economically important feature of the Uruguayan retail sector and, as described below, motivates our distinction between physical stores and economic sellers.

The database contains daily prices from April 1, 2007, to December 31, 2024, for 154 products, the vast majority of which are identified at the UPC level. This level of detail allows us to track identical goods across stores and locations over time, avoiding measurement problems generated by product heterogeneity (Atkin and Donaldson, 2015). The products in the full reporting system represent 15.6% of the Uruguayan Consumer Price Index (CPI) basket. Product definitions specify precise varieties and presentations to make observations comparable across stores. For example, stores reporting Coca-Cola

¹This database updates and extends versions used in Borraz and Zipitría (2012), Borraz, Cavallo, Rigobon, and Zipitría (2016), Borraz and Saldain (2017), Borraz and Zipitría (2022), and Kłaczko (2025).

carbonated soft drinks report the specified 1.5-liter presentation; no price is recorded when that variety is unavailable.

Products were originally selected from a 2006 survey of major chains, generally using the three best-selling brands in each category, and the monitored list underwent a major revision in November 2010. The full DGC list contains 154 products in more than 60 categories. We retain 125 products that can be consistently identified over time, grouped into 42 consistently defined categories. We exclude unpackaged goods such as ham, meat, and poultry, as well as drugstore items for which product coverage is incomplete. The resulting raw sample contains nearly 60 million daily price observations. We remove extreme observations defined as prices more than three times or less than one-third of the product-specific monthly median; these account for less than 0.01% of observations. The complete product list is reported in Online Appendix [D](#).

We compute a monthly modal posted price for each product–store pair to attenuate short-lived promotions, motivated by evidence that sales account for substantial price variation while reference prices are more inertial (Nakamura, Nakamura, and Nakamura, 2011; Eichenbaum, Jaimovich, and Rebelo, 2011). Monthly aggregation retains within-month price differences but does not produce a formal regular-price series: prolonged promotions, stale reports, and reporting asynchrony can remain (Sheremirov, 2020; Glendon, 2018).

We deflate monthly modal prices by the CPI and conduct the analysis using their logarithms. The final monthly store–product panel contains approximately 3.2 million price observations.

The data include detailed store information, including exact geographic location (Universal Transverse Mercator coordinates) and chain affiliation. Although the reporting system covers stores throughout Uruguay, most cities outside Montevideo contain too few simultaneously observed sellers to support consistent local same-good price comparisons over a long horizon. We therefore focus on Montevideo, Uruguay’s capital and largest city, which accounts for nearly one-third of the country’s population and 54% of stores

in the reporting system. We define local markets at the neighborhood level. The resulting unbalanced panel contains 61 local markets and 324 stores observed at some point between 2007 and 2024.

For baseline local dispersion, we retain physical stores and compare CPI-adjusted log prices for the same product within market and month, requiring at least two reporting stores; Online Appendix Section A.1 documents the resulting unbalanced support. For across-seller analyses, let p_{ijmt} denote the log CPI-adjusted monthly modal price of product i at physical store j in market m and month t . An *economic seller* g is either a retail chain or an independent store. Let $\mathcal{J}_{imt}^g$ be the branches of seller g reporting product i in that market and month, with a singleton set for an independent store, and define

$$\tilde{p}_{igmt} = \text{median}_{j \in \mathcal{J}_{imt}^g} p_{ijmt}.$$

This economic-seller price is used throughout the across-seller analyses; product–market–month cells require at least two economic sellers.

The DGC data report posted prices rather than transaction quantities. Accordingly, our measures characterize the cross-sectional distribution of available seller prices rather than an expenditure-weighted distribution of transactions. This feature is especially relevant for the decomposition in Section 4: seller-wide price positions are constructed from the comparable products observed for each seller, without sales weights. We return to product coverage and seller composition in the robustness analysis.

Table 1 summarizes the monthly price panel and the physical-store and economic-seller comparison samples.

Table 1: Summary Statistics.

	Observations	Mean	Std. Dev.	Median
<i>Panel A. Monthly store–product price data</i>				
CPI-adjusted log price	3,257,937	3.283	0.548	3.257
<i>Panel B. Physical-store product–market–month comparisons</i>				
Price dispersion (%)	771,006	5.553	6.697	3.787
Average CPI-adjusted log price	771,006	3.274	0.544	3.245
Stores per product–market–month	771,006	3.743	3.032	3.000
<i>Panel C. Across-economic-seller product–market–month comparisons</i>				
Across-seller price dispersion (%)	648,399	6.627	7.066	4.816
Average CPI-adjusted log price	648,399	3.282	0.545	3.255
Economic sellers per product–market–month	648,399	3.027	1.472	3.000
<i>Panel D. Sample coverage</i>				
Sample period			04/2007–12/2024	
Monthly price observations				3,257,937
Stores				324
Products				125
Markets				61
Categories				42
Chains				22

Notes: Panel A summarizes the monthly store–product data. Panels B and C require at least two physical stores or economic sellers, respectively; dispersion is $100 \times SD(\log p)$. Economic sellers are defined in Section 2. Panel D reports sample coverage; Online Appendix Section A.1 documents comparison-cell size.

3 Long-Run Price Dispersion

This section establishes three empirical facts. Same-good local price dispersion rises substantially over the sample, with an acceleration after the mid-2010s and a partial reversal after 2021–2022. Average real prices follow a different path and inflation controls leave the fitted widening largely unchanged. The increase occurs across economically distinct sellers—including among retail chains—rather than across branches of the same chain.

3.1 A Long-Run Increase in Local Price Dispersion

Following Sheremirov (2020), we measure price dispersion for each product i , market m , and month t as the cross-store standard deviation of log prices,

$$SD_{imt} = 100 \times sd_s(p_{ismt}),$$

where p_{ismt} is the log CPI-adjusted price charged by store s . Deflation makes price levels comparable in real terms over time. Because the monthly CPI is common to all stores in a product–market–month cell, subtracting its logarithm leaves the cross-store standard deviation unchanged. We also compute $\tilde{p}_{imt}$, the corresponding cross-store average log price. We report 100 times the standard deviation of log prices; one unit therefore corresponds to one percentage point of the standard deviation of log prices. Changes in this measure are reported in percentage points, and its variance components in squared percentage points. Thus, SD_{imt} measures price dispersion for the same product in the same local market and month.

Borrowing from Dvir and Strasser (2018), we study the evolution of price dispersion using the average log price, a time trend, and fixed effects. We adapt their approach to local product-level comparisons and use a linear monthly trend as a summary of the time path. Specifically, we estimate variants of

$$SD_{imt} = \alpha + \beta \tilde{p}_{imt} + \gamma t + \delta' FE_{imt} + \varepsilon_{imt}, \quad (1)$$

where t is a linear monthly time trend and FE_{imt} denotes alternative sets of fixed effects. The coefficient γ measures the average evolution of local price dispersion under increasingly demanding within-product and within-market comparisons. All specifications are estimated by weighted least squares, using the number of stores underlying each dispersion measure as weights. Standard errors are clustered at the product-category, market, and time levels.

Monthly trend estimates range from 0.0164 to 0.0167 percentage points; the preferred product-by-market specification gives 0.0166, implying a fitted increase of about 3.5 percentage points over the sample. The endpoint comparison is smaller: average dispersion rises from 4.76 to 6.82 on the $100 \times SD(\log p)$ scale, while across-seller dispersion rises by

2.29 percentage points. Because the series peaks around 2021–2022 and partially reverses, the linear coefficient summarizes the low-frequency movement rather than a monotonic path. Product-by-market fixed effects absorb persistent cell differences; composition is examined in Section 3.4.

Table 2: Baseline Price-Dispersion Trend Estimates.

Dependent Variable: Model:	(1)	(2)	SD (in %)		(5)	(6)
			(3)	(4)		
<i>Variables</i>						
Av. Price	-2.087*** (0.4969)	-7.359*** (1.542)	-6.864*** (1.460)	-6.853*** (1.461)	-5.994*** (1.286)	-5.981*** (1.287)
Time	0.0166*** (0.0020)	0.0164*** (0.0022)	0.0167*** (0.0021)	0.0167*** (0.0021)	0.0166*** (0.0020)	0.0166*** (0.0020)
<i>Fixed-effects</i>						
Market	Yes		Yes	Yes		
Product		Yes	Yes	Yes		
Month				Yes		Yes
Product $\times$ Market					Yes	Yes
<i>Fit statistics</i>						
Observations	771,006	771,006	771,006	771,006	770,942	770,942
R ²	0.14453	0.17624	0.26022	0.26049	0.37652	0.37679
Within R ²	0.06258	0.04869	0.05031	0.05029	0.05029	0.05028

Clustered (Category & Market & Time) standard-errors in parentheses

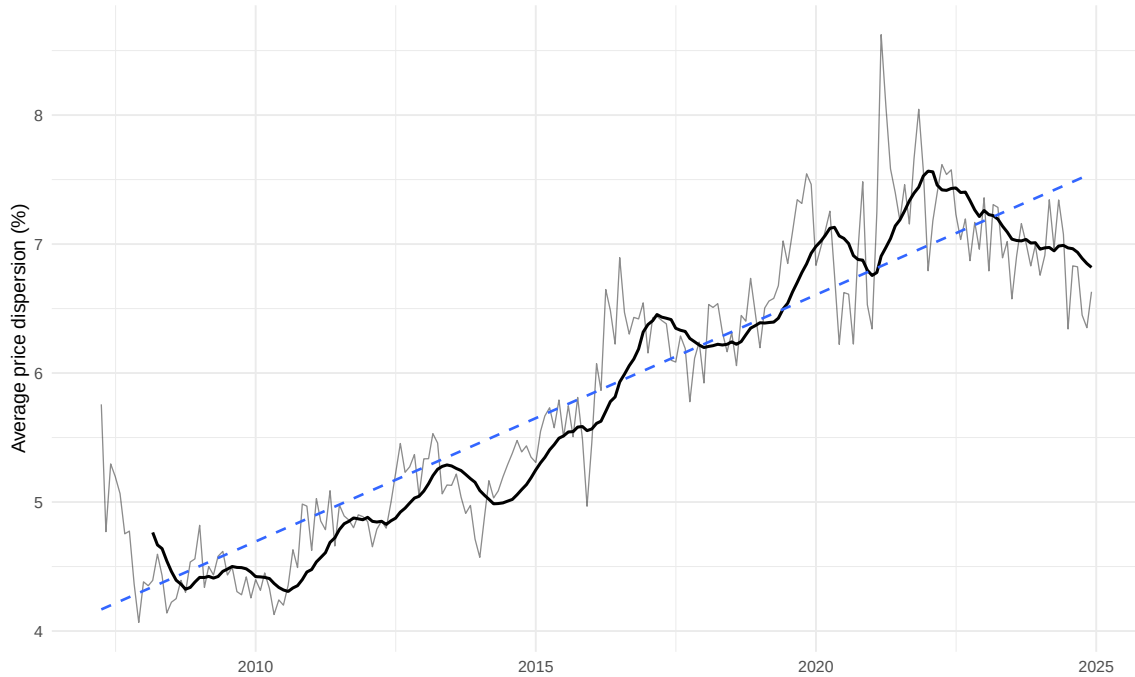
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is $100 \times SD(\log p)$ across physical stores within product–market–month. *Av. Price* is the corresponding mean log CPI-adjusted price and *Time* a linear monthly trend. Fixed effects are indicated in the table. Regressions use store-count weights and three-way clustered standard errors by product category, market, and time.

Figure 1 shows the same non-monotonic long-run pattern.

Klaczko (2025) documents increasing dispersion over an earlier portion of the same reporting system. The longer panel establishes that the widening persists over nearly eighteen years and allows us to study how its composition changes. Flexible year effects in Online Appendix Figure A.2 reproduce the widening and later partial reversal without imposing a linear path.

Figure 1: Price Dispersion over Time.



Notes: Weighted monthly mean of the baseline dispersion measure in Section 3.1. Thin line: monthly series; thick line: 12-month moving average; dashed line: linear fit.

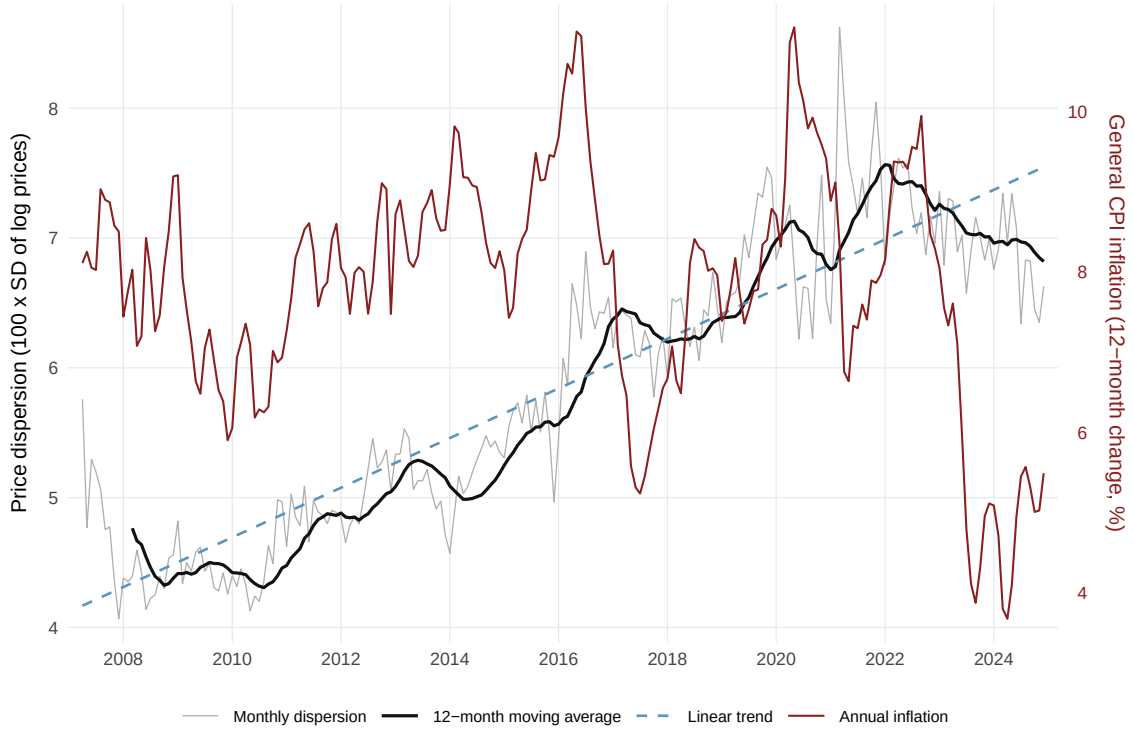
3.2 Average Real Prices and Inflation

Average real prices display substantial medium-run movements rather than the sustained widening of dispersion. Flexible monthly effects and non-overlapping April–March blocks in Online Appendix Figures A.3 and A.4 show a decline around 2018–2020 followed by recovery. The comparison concerns the different paths of the distribution’s center and spread; a common monthly CPI deflator leaves cross-sectional log-price dispersion unchanged.

Figure 2 shows no stable one-for-one relation between dispersion and twelve-month general CPI inflation: inflation falls sharply after its 2016 peak while dispersion remains elevated, and the two series only partly comove near the end of the sample.

The comparison complements work on dispersion across inflation regimes, price concepts, and product-specific inflation (Álvarez, Beraja, González-Rozada, and Neumeyer, 2019; Sheremirov, 2020; Caglayan, Filiztekin, and Rauh, 2008). Across controls for current, lagged, cumulative, and volatile aggregate inflation, the fitted dispersion trend

Figure 2: Same-Good Price Dispersion and General CPI Inflation



Notes: Left axis: the weighted monthly baseline dispersion measure from Section 3.1; gray line: monthly series, black line: 12-month moving average, dashed line: descriptive linear fit. Right axis: 12-month general CPI inflation, $100(CPI_t/CPI_{t-12} - 1)$. The axes have separate scales; the fitted line is not the conditional trend in Table 2.

changes by at most 3.4 percent relative to the corresponding same-sample baseline. Using matched category inflation yields a small and statistically insignificant contemporaneous coefficient (Online Appendix Sections A.3–A.4).

3.3 Across-Seller and Within-Chain Dispersion

We next locate the widening within the retail organization. Previous Uruguayan evidence documents low within-chain dispersion and an important chain component, while international work similarly finds centralized or zone-based chain pricing (Borraz and Saldain, 2017; Klaczko, 2025; DellaVigna and Gentzkow, 2019; Adams and Williams, 2019; Hitsch, Hortag su, and Lin, 2021). We measure within-chain dispersion across branches of the same chain and across-seller dispersion after collapsing each chain to a single economic seller; independent stores remain separate sellers.

For within-chain dispersion, we restrict attention to retail chains g and use the physical-store prices within the branch set $\mathcal{J}_{imt}^g$ defined in Section 2:

$$SD_{imt}^{\text{within}} = 100 \times sd_{j \in \mathcal{J}_{imt}^g}(p_{ijmt}).$$

The unit of observation is product–market–month–chain.

Let $\mathcal{G}_{imt}$ denote the economic sellers observed for product i in market m and month t . Using the economic-seller price $\tilde{p}_{igmt}$ defined in Section 2, across-seller dispersion is

$$SD_{imt}^{\text{seller}} = 100 \times sd_{g \in \mathcal{G}_{imt}}(\tilde{p}_{igmt}).$$

This measure removes within-chain branch variation and isolates price differences across economically distinct pricing units in the same local product market. It is the starting point for the decomposition in the next section.

The contrast is sharp: across-seller dispersion rises by about 0.019 percentage points per month, whereas the within-chain trend is close to zero. Average within-chain dispersion is also much lower (0.96 versus 6.63 on the respective supports). The long-run widening therefore occurs across economic sellers rather than through increasing differentiation across branches of the same chain.

Figure 3 shows the same divergence in the monthly series.

Table 3: Price Dispersion Trends Across Sellers and Within Chains.

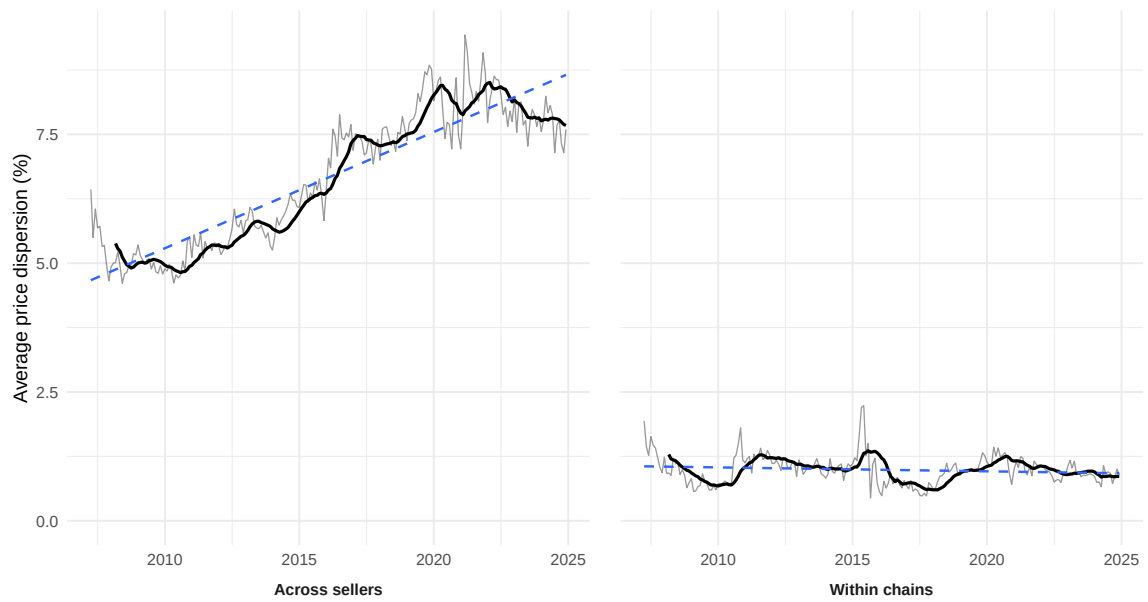
Dependent Variable:	SD (in %)					
	Across sellers			Within chains		
Model:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Av. Price	-9.451*** (1.747)	-9.107*** (1.713)	-7.792*** (1.521)	-2.219*** (0.5848)	-1.634*** (0.4590)	-1.542*** (0.4516)
Time	0.0193*** (0.0025)	0.0193*** (0.0024)	0.0192*** (0.0023)	-0.0002 (0.0014)	0.0005 (0.0015)	0.0006 (0.0015)
<i>Fixed-effects</i>						
Product	Yes	Yes		Yes	Yes	
Market		Yes			Yes	
Product $\times$ Market			Yes			Yes
<i>Fit statistics</i>						
Observations	648,399	648,399	648,318	513,346	513,346	513,319
R ²	0.21759	0.24032	0.36553	0.04086	0.13238	0.23111
Within R ²	0.06536	0.06319	0.06089	0.00996	0.00583	0.00533

Clustered (Category & Market & Time) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Columns (1)–(3) use across-seller dispersion and columns (4)–(6) within-chain dispersion, both defined in Section 3.3; economic sellers are defined in Section 2. Regressions use seller-count or store-count weights, respectively, and three-way clustered standard errors by product category, market, and time.

Figure 3: Price Dispersion over Time: Across Sellers and Within Chains.



Notes: Weighted monthly means of the across-seller and within-chain measures defined in Section 3.3. Thin lines: monthly series; thick lines: 12-month moving averages; dashed lines: linear fits.

3.4 Divergence Across Retail Chains

The across-seller results establish that the secular increase in local price dispersion occurs across economically distinct sellers rather than across branches belonging to the same retail chain. The across-seller measure, however, combines two different types of comparisons: price differences among retail chains and price differences between chains and independent sellers. We therefore separate these margins to determine whether the long-run increase is driven by independent retailers or is already present among retail chains themselves.

Using the economic-seller prices defined in Section 2, we divide observed seller pairs within each product–market–month into chain–chain and chain–independent comparisons.

To express the two margins on a common scale, for pair type $r \in \{CC, CI\}$ we define

$$D_{imt}^r = 100 \sqrt{\frac{1}{2|\mathcal{P}_{imt}^r|} \sum_{(g,h) \in \mathcal{P}_{imt}^r} (\tilde{p}_{igmt} - \tilde{p}_{ihmt})^2}, \quad (2)$$

where $\mathcal{P}_{imt}^{CC}$ contains all pairs of distinct retail chains and $\mathcal{P}_{imt}^{CI}$ contains all chain–independent pairs observed for product i , market m , and month t . For chain–chain comparisons, this measure is numerically equivalent to the sample standard deviation of chain prices. The same normalization places chain–independent price gaps on the corresponding standard-deviation scale.

Because the availability of the two types of comparisons differs across product–market–month observations, our preferred comparison uses their common support: observations for which both a chain–chain and a chain–independent measure can be constructed. We stack the two observations within each product–market–month and estimate a specification that allows product-by-market fixed effects and the coefficient on average price to differ by pair type. The coefficient on *Time* identifies the chain–chain trend, while its interaction with the chain–independent indicator identifies the difference between the two trends.

Table 4: Long-Run Price-Dispersion Trends by Seller Pair Type.

Comparison	Monthly trend	Std. Error	p -value	Cumulative change (pp)
Chain–Chain	0.0202***	0.0027	< 0.001	4.29
Chain–Independent	0.0194***	0.0027	< 0.001	4.11
Difference: CI – CC	-0.0008	0.0023	0.715	-0.18

Notes: Common-support estimates for the pair-type dispersion in Equation (2). The stacked regression allows product-by-market fixed effects and the average-price slope to differ by pair type, uses seller-count weights, and clusters standard errors by product category, market, and time. The difference is estimated in the stacked sample; cumulative changes equal the monthly trend times 212. Significance: *** 1%, ** 5%, * 10%.

On common support, chain–chain and chain–independent dispersion rise at nearly the same rate (0.0202 and 0.0194 percentage points per month), with an estimated difference of -0.0008 ($p = 0.715$). Widening is therefore already present in chain-level relative prices and is not confined to independents moving away from chains. Online Appendix Section A.7 reports full-support comparisons.

Robustness. The widening also survives restrictions on seller and product composition, fixed seller counts, Newey–West inference on monthly aggregates, exclusion of 2020–2021, and leave-one-chain-out checks. Online Appendix Sections A.3 and A.5–A.7 report these exercises.

4 The Changing Structure of Relative-Price Dispersion

The long-run widening can reflect changes in sellers’ general price positions or in the relative positions of their individual products. Using the construction of Kaplan, Menzio, Rudanko, and Trachter (2019), with Kaplan and Menzio (2015) as its precursor, we attribute the widening to seller-wide and seller-product positions and then distinguish category-wide from within-category contributions. Retail organization defines the sellers compared throughout.

4.1 Price Components and Their Long-Run Trends

Following the initial price-normalization and component-construction steps of Kaplan, Menzio, Rudanko, and Trachter (2019), we normalize prices and separate seller-wide from seller-product positions. We adapt the construction to the economic-seller price $\tilde{p}_{igmt}$ defined in Section 2. We first remove the contemporaneous local price of each exact product,

$$r_{igmt} = 100 \left(\tilde{p}_{igmt} - \bar{\tilde{p}}_{imt} \right), \quad (3)$$

where $\bar{\tilde{p}}_{imt}$ is the mean log price across economic sellers observed for product i in market m and month t . Thus, r_{igmt} measures how expensive or cheap seller g is for product i relative to competing sellers offering that same product in the same local market and month. By construction, common product-level price movements cancel from this relative-price measure. This normalization does not remove a possible influence of inflation on differences across sellers: inflation may still be associated with changes in the timing, magnitude, or persistence of their relative-price adjustments.

We then summarize seller g 's general price position in market m and month t by averaging its relative prices over the comparable products it carries. Let $\mathcal{I}_{gmt}$ denote the set of products carried by seller g for which a same-product comparison with at least one other economic seller is available. We define the seller component as

$$S_{gmt} = \frac{1}{|\mathcal{I}_{gmt}|} \sum_{i \in \mathcal{I}_{gmt}} r_{igmt}. \quad (4)$$

A positive S_{gmt} means that seller g is, on average, relatively expensive across the comparable products it sells in that market and month; a negative value means that it is relatively cheap. This is the analogue of the store component in Kaplan, Menzio, Rudanko, and Trachter (2019), constructed here for economic sellers within local markets.

The residual seller-product component, which can include both category-wide and within-category price positions, is

$$G_{igmt} = r_{igmt} - S_{gmt}. \quad (5)$$

A positive G_{igmt} means that product i is unusually expensive at seller g relative to that seller's overall local price position. Conversely, a negative value indicates that the product is relatively cheap at that seller even after accounting for whether the seller is generally expensive or cheap. Equations (4) and (5) imply the exact identity

$$r_{igmt} = S_{gmt} + G_{igmt}. \quad (6)$$

The distinction is economically important. If sellers become increasingly separated into generally high- and low-price retailers, dispersion in S_{gmt} should rise. If instead the widening occurs because sellers increasingly charge different relative prices product by product, dispersion in G_{igmt} should rise even if their average price positions remain similar.

For any exact additive decomposition $x = a + b$, we use the variance accounting identity

$$Var(x) = Var(a) + Var(b) + 2Cov(a, b).$$

We apply this identity here and again to the reset/erosion decomposition in Section 5. Within each product-market-month, define

$$\begin{aligned} V_{imt}^T &= Var_g(r_{igmt}), & V_{imt}^S &= Var_g(S_{gmt}), \\ V_{imt}^{SG} &= Var_g(G_{igmt}), & C_{imt} &= 2Cov_g(S_{gmt}, G_{igmt}). \end{aligned}$$

Demeaning prices within product-market-month leaves their cross-seller variance unchanged, so V_{imt}^T corresponds exactly to the across-seller dispersion measure in Section 3. Applying the identity to Equation (6) gives

$$V_{imt}^T = V_{imt}^S + V_{imt}^{SG} + C_{imt}. \quad (7)$$

All terms are in squared percentage-point units; the covariance records comovement between seller-wide and seller-product positions rather than a separate mechanism.

Our trend-accounting exercise applies the specification in Equation (1) to each term of Equation (7). For each component $k \in \{T, S, SG, C\}$, we estimate

$$V_{imt}^k = \alpha_k + \beta_k \tilde{p}_{imt} + \gamma_k t + \eta_{k,im} + \lambda_{k,c(t)} + u_{imt}^k, \quad (8)$$

where $\eta_{k,im}$ denotes product-by-market fixed effects and $\lambda_{k,c(t)}$ calendar-month fixed effects, with $c(t)$ denoting the calendar month corresponding to observation t . The regressions use the same product–market–month observations for every component, are weighted by the number of economic sellers underlying the variance, and use standard errors three-way clustered by product category, market, and time. Because Equation (7) holds observation by observation and the four regressions use identical samples, regressors, weights, and fixed effects, the estimated trend coefficients also satisfy

$$\hat{\gamma}_T = \hat{\gamma}_S + \hat{\gamma}_{SG} + \hat{\gamma}_C$$

up to numerical precision. Throughout the remainder of the empirical analysis, unless a statistic is explicitly described as an endpoint change, a component’s *share* or *contribution* is its fitted time-trend coefficient divided by the relevant parent trend. These are not shares of variance levels or of the observed first-to-last change, and they are distinct from KMRT’s fixed, persistent, and transitory statistical components.

4.2 Which Component Is Widening?

Table 5 reports fitted component trends together with first- and last-12-month averages.

Seller-product positions account for 90 percent of the fitted increase in total variance and 79.1 percent of the increase between the first and final twelve months. The widening therefore mainly reflects divergence in sellers’ relative-price profiles rather than increasing differences in their overall price levels. In levels, the seller-wide component rises only from

Table 5: Dynamic Decomposition of Across-Seller Price Variance.

Component	Monthly trend	Std. Error	Fitted cumulative change	First 12m avg.	Last 12m avg.
Total variance	0.4971***	0.0826	105.39	57.15	101.15
Seller component	0.0440***	0.0071	9.32	6.48	13.86
Seller-product component	0.4480***	0.0780	94.98	50.24	85.06
$2 \times$ covariance	0.0051	0.0093	1.09	0.43	2.23

Notes: Components are defined in Section 4.1; all variance terms are in squared percentage-point units. Trend estimates use Equation (8). Fitted cumulative change is the monthly trend times 212; first- and last-12-month columns are weighted averages of the raw component series. Significance: *** 1%, ** 5%, * 10%.

11 to 14 percent of across-seller variance, close to the 10 and 15 percent reported for the United States by Kaplan and Menzio (2015) and Kaplan, Menzio, Rudanko, and Trachter (2019). The comparison is indicative because those studies use weekly transaction prices whereas we use monthly modal posted prices; the similarity concerns the order of magnitude of the seller-wide share rather than exact comparability. While those studies document the composition of price dispersion, the long panel allows us to identify which component drives its evolution over time.

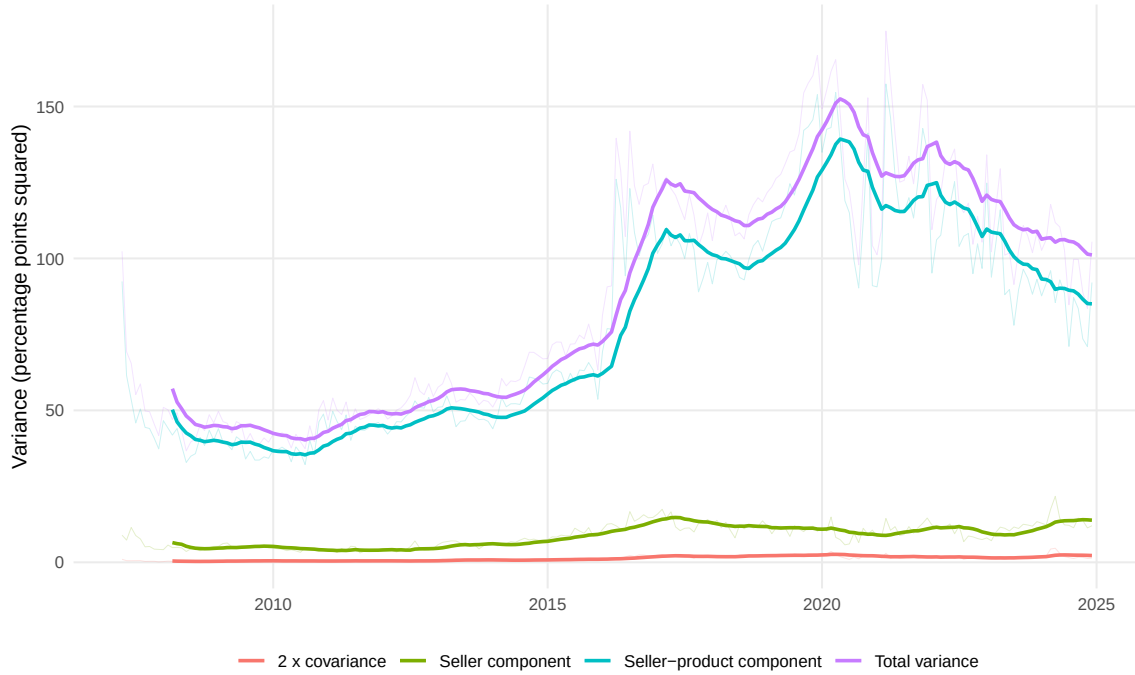
Figure 4 shows that the seller-product component tracks the widening and subsequent compression of total variance, while the seller-wide component remains much smaller.

4.3 Category-Wide and Within-Category Positions

The seller-product component can reflect broad category positions as well as product-specific departures within categories. We therefore define a seller-category benchmark as the seller’s average residual position across comparable products in a category and a within-category component as each product’s departure from that benchmark.

In the full sample, category-wide and within-category positions each contribute about 46 percent of the seller-product trend. Restricting category benchmarks to seller-category-market-month groups with at least two comparable products raises the within-category share to 56 percent, versus 40 percent for category-wide positions. The widening therefore reflects both margins and, on multiproduct support, somewhat more divergence among

Figure 4: The Composition of Across-Seller Price Variance over Time.



Notes: Weighted monthly means of the variance components defined in Section 4.1, in squared percentage-point units. Thin lines show monthly values and thick lines 12-month moving averages.

individual products within categories. Online Appendix Section A.8 reports the construction and estimates. This extends the category-versus-product distinction in retail price dynamics (Nakamura, Nakamura, and Nakamura, 2011) to contributions to the long-run variance trend.

4.4 Robustness of the Variance Decomposition

The seller-product dominance is stable to an alternative fixed-effect seller benchmark, a long-lived-product sample, and tail restrictions: its fitted share remains about 90 percent under the first two exercises and between 86.8 and 89.4 percent after trimming or winsorizing extremes. Online Appendix Sections A.9–A.10 report these checks.

5 Price Adjustment and Reset-Price Dispersion

The decomposition locates the widening in the cross section. We now ask how much remains after reversing CPI erosion since each price’s last observed nominal adjustment. This links relative-price positions to the price-setting literature on frequency, duration, adjustment size, and temporary sales (Klenow and Kryvtsov, 2008; Nakamura and Steinsson, 2008; Klenow and Malin, 2011). Dispersion can be present at young prices (Campbell and Eden, 2014); the question here is its contribution to the long-run seller-product trend. Unlike reset-price inflation as a measure of desired-price changes (Bils, Klenow, and Malin, 2012), our reconstruction uses historical observed resets.

5.1 Reconstructing Prices at the Last Observed Adjustment

We use nominal monthly modal posted prices to separate these accounting contributions. Let p_{igmt}^N denote the log nominal counterpart of the economic-seller price $\tilde{p}_{igmt}$ defined in Section 2, and let P_t denote the log CPI. Let $\tau_{igmt} \leq t$ be the month of the most recent *observed* nominal-price change for that seller-product-market observation. An observed reset is therefore a change in the monthly modal nominal price, not the exact date of a daily repricing decision. For observations for which the previous reset is observed, the current real log price can be written exactly as

$$p_{igmt}^A = p_{igmt\tau_{igmt}}^N - P_t = \underbrace{p_{igmt\tau_{igmt}}^N - P_{\tau_{igmt}}}_{p_{igmt}^R} - \underbrace{(P_t - P_{\tau_{igmt}})}_{E_{igmt}}, \quad (9)$$

where p_{igmt}^A is the actual real price, p_{igmt}^R is the real price at the most recent observed reset, and E_{igmt} is cumulative inflation since that reset. Equation (9) is an accounting identity. It does not assume Calvo adjustment, a particular menu cost, or that the observed reset price equals a frictionless desired price.

We apply to p_{igmt}^A , p_{igmt}^R , and E_{igmt} the same linear transformations used in Section 4.1, including the factor of 100. The comparison sets and weights are identical for all three

terms at each date. Within each product–market–month we remove the contemporaneous seller mean, and within each seller–market–month we remove the seller-wide position. Let the resulting seller-product components be G_{igmt}^A , G_{igmt}^R , and G_{igmt}^E . Linearity implies

$$G_{igmt}^A = G_{igmt}^R - G_{igmt}^E. \quad (10)$$

G^R is reconstructed on date- t comparison sets from each observation’s most recent reset, so reset dates differ across sellers and longer spells receive more representation. The exercise measures how much current seller-product widening survives reversing CPI erosion on observed spells; it is not a distribution of desired prices or reset events. Seller-wide normalized erosion is removed by construction.

Applying the variance accounting identity from Section 4.1 to Equation (10), define

$$V_{imt}^{NE} \equiv V_{imt}^E - 2 \text{Cov}_g(G_{igmt}^R, G_{igmt}^E), \quad V_{imt}^A = V_{imt}^R + V_{imt}^{NE}. \quad (11)$$

The net term combines erosion variance with its covariance with reset positions and is the relevant contribution to the observed seller-product trend.

5.2 How Much Widening Remains in the Last-Reset Reconstruction?

We use the specification, weights, and inference of Equation (8). Table 6 shows that, on spells with observed beginnings, 95.2 percent of the seller-product variance trend remains in the last-reset reconstruction. The positive contribution of erosion variance is largely offset by its covariance with reset positions: the net contribution is 4.8 percent and imprecisely estimated ($p = 0.472$). Thus, the widening on this support is largely retained after reversing subsequent CPI erosion.

Figure 5 shows reset-position variance tracking observed G over time, while the net erosion contribution remains much smaller. Unlike the aggregate inflation controls in

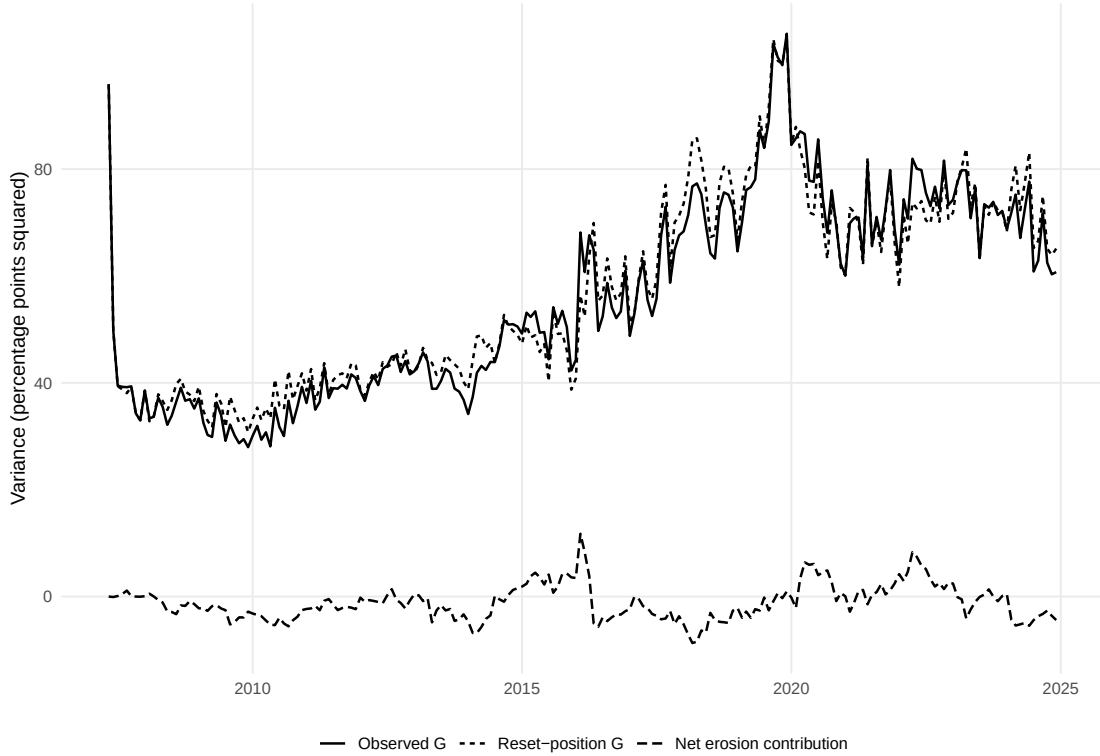
Table 6: Reset-Price Positions and Inflation Erosion

Component	Monthly trend	Std. Error	p -value	Fitted change	Contribution (%)
Observed seller-product variance	0.2729***	0.0494	< 0.001	57.59	100.0
Reset-price position	0.2598***	0.0490	< 0.001	54.81	95.2
Inflation erosion	0.0589***	0.0170	0.001	12.42	21.6
Reset/erosion covariance	-0.0457***	0.0156	0.005	-9.65	-16.8
Net erosion contribution	0.0131	0.0181	0.472	2.77	4.8

Notes: Components and the observed-spell support are defined in Section 5. Trend estimates use Equation (8). Fitted change is the monthly trend accumulated over the identified-reset sample; contribution divides each component trend by the observed G trend, following the convention in Section 4.1. Significance: *** 1%, ** 5%, * 10%.

Section 3.2, this accounting exercise uses each price's own observed reset history.

Figure 5: Seller-Product Dispersion at Reset and Between Resets.



Notes: Seller-count-weighted monthly series on the observed-spell support. Observed G , reset-position G , and net erosion are defined in Section 5.

5.3 Support, Robustness, and Adjustment Timing

The identified-spell support has a seller-product trend equal to about 61 percent of the full-sample trend, reflecting both cell and within-cell seller selection; the 95.2 percent reset-position share therefore applies to this restricted support. Its internal allocation is nevertheless stable across alternative spell constructions, with reset shares between 94.8 and 98.8 percent (Online Appendix Sections [A.11–A.12](#)).

Adjustment timing changes along several margins. Monthly reset frequency averages 25.9 percent and has no significant trend over 2008–2024, while conditional changes become larger and the upper tail of observed price age lengthens; from 2012 onward, reset frequency rises even as the long-duration tail expands. Online Appendix Section [A.13](#) reports these moments. The model below illustrates why the organization of price gaps within firms can matter independently of a single timing statistic.

6 Monetary Transmission: A Multiproduct Illustration

The empirical evidence documents how the composition of same-good price dispersion changes over time. This section illustrates why that composition can matter for monetary transmission. A multiproduct firm’s inherited price gaps can be aligned across products or have opposite signs, even when every product has the same cross-sectional price distribution in the two economies. With economies of scope in repricing, these configurations can lead to different adjustment decisions following the same nominal expenditure shock when the common cost is large enough that two large gaps jointly activate repricing while one large gap alone does not. A relative-shock experiment establishes the boundary of this result: relative shocks can induce universal adjustment and full nominal pass-through. The model illustrates possible effects of the organization of price gaps; it is not an estimated explanation of the observed dispersion trend.

The pricing technology is the finite two-product counterpart of Bonomo, Carvalho, Kryvtsov, Ribon, and Rigato (2023). A firm that changes at least one price pays a common fixed cost, and it also pays an individual cost for each product repriced. This nests the full-synchronization logic of Alvarez and Lippi (2014) as the limiting case in which the individual cost is zero, while allowing partial synchronization when both costs are positive. We combine this adjustment technology with the nominal-expenditure environment used below to isolate how the within-firm organization of gaps affects selection.

6.1 Environment and Price Choice

There is a representative household, two products, and a unit mass of atomistic sellers, each offering both products. Preferences are $\log C - \chi N$, where $C = 2\sqrt{C_1 C_2}$ and each C_i is a CES aggregate across sellers with elasticity $\varepsilon > 1$. Differentiation represents retail services associated with otherwise comparable physical goods. Thus, sellers have market power; this is not homogeneous-good Bertrand competition. The corresponding price indices are

$$P_i = \left(\int_0^1 p_{gi}^{1-\varepsilon} dg \right)^{1/(1-\varepsilon)}, \quad P = \sqrt{P_1 P_2}. \quad (12)$$

The household owns firms and satisfies $PC = WN + \Pi$, with profits net of adjustment costs. The labor condition is $W/P = \chi C$. We close aggregate nominal expenditure at $PC = M$, obtaining

$$W = \chi M, \quad C = M/P, \quad c_{gi} = \frac{M}{2} P_i^{\varepsilon-1} p_{gi}^{-\varepsilon}. \quad (13)$$

Here M is a nominal expenditure anchor. A nominal expenditure shock can be interpreted as a monetary shock under fixed velocity; no interest-rate rule or money-demand equation is derived. The model describes impact responses in a temporary equilibrium with inherited prices.

Production is $y_{gi} = A_{gi} n_{gi}$. Let $J_g \subseteq \{1, 2\}$ denote the subset of products repriced by firm g , and let $m(J_g) = |J_g|/2$ be the fraction of its two products that change. Following

the adjustment-cost structure of Bonomo, Carvalho, Kryvtsov, Ribon, and Rigato (2023), repricing uses

$$\kappa(J_g) = K \mathbf{1}\{J_g \neq \emptyset\} + c m(J_g) = K \mathbf{1}\{J_g \neq \emptyset\} + \frac{c}{2}|J_g| \quad (14)$$

units of labor. Thus a one-product change costs $K + c/2$, whereas repricing both products costs $K + c$. The common cost K creates economies of scope; the individual cost c makes selective repricing possible.

Taking aggregate indices as given, a firm's operating profit is

$$\pi_g(\mathbf{p}) = \frac{M}{2} \sum_{i=1}^2 P_i^{\varepsilon-1} \left[p_{gi}^{1-\varepsilon} - \frac{W}{A_{gi}} p_{gi}^{-\varepsilon} \right]. \quad (15)$$

Conditional on repricing product i , its unique optimal reset price is

$$p_{gi}^* = \frac{\varepsilon}{\varepsilon - 1} \frac{W}{A_{gi}}. \quad (16)$$

For any candidate subset J , let $\mathbf{p}_g(J)$ set $p_{gi} = p_{gi}^*$ for $i \in J$ and retain the inherited price p_{gi}^0 otherwise. The firm chooses

$$J_g^* \in \arg \max_{J \subseteq \{1,2\}} \{ \pi_g(\mathbf{p}_g(J)) - W \kappa(J) \}. \quad (17)$$

At equality we select inaction; all numerical decisions below are strict. Price indices must be consistent with firms' choices. Total labor includes production and repricing costs, so the resource constraint and the household budget hold jointly. Unlike the pure joint-cost benchmark, the firm is free to reprice neither product, either single product, or both.

6.2 Identical Product Distributions, Different Nominal Transmission

Let $x_{gi} = \log p_{gi}^0 - \log p_{gi}^*$ denote an inherited price gap. Define

$$a_g = (x_{g1} + x_{g2})/2, \quad d_g = (x_{g1} - x_{g2})/2. \quad (18)$$

The common (absolute) coordinate a_g shifts both product gaps together; d_g shifts them in opposite directions. These are coordinates of *gaps*, not realized price changes or product-level means across sellers. The empirical G component measures observed relative-price positions, whereas d_g measures a gap from desired relative prices. The two share a geometry, not an identity. In particular, if $q_g = (\log p_{g1} - \log p_{g2})/2$ and $q_g^* = (\log p_{g1}^* - \log p_{g2}^*)/2$, then $d_g = q_g^0 - q_g^*$. General seller-level price movements are also distinct from aggregate inflation.

Normalize $M_0 = A_{gi} = 1$ and $\chi = (\varepsilon - 1)/\varepsilon$, so initial desired prices equal one. Around symmetry, the gross labor-unit gain from resetting product i is

$$q(x_{gi}) = \frac{\varepsilon}{4} x_{gi}^2. \quad (19)$$

Conditional on paying the common cost, product i is worth repricing when $q(x_{gi}) > c/2$. The firm acts when the sum of the positive product-level surpluses exceeds the common cost:

$$\sum_{i=1}^2 \left[q(x_{gi}) - \frac{c}{2} \right]_+ > K. \quad (20)$$

This is the two-product analogue of the partial-synchronization mechanism in Bonomo, Carvalho, Kryvtsov, Ribon, and Rigato (2023): K links adjustment decisions across products, while c allows the firm to leave small gaps unchanged even after repricing activity begins.

A nominal expenditure shock $\delta = \Delta \log M$ shifts gaps from (a_g, d_g) to $(a_g - \delta, d_g)$.

Consider two equally weighted types of firms in each economy:

	Type 1 (a, d)	Type 2 (a, d)	
Common gaps (C)	$(-r, 0)$	$(r, 0)$	(21)
Relative gaps (R)	$(0, -r)$	$(0, r)$	

In both economies, *each product across sellers* has half its prices at e^{-r} and half at e^r . Initial product means, variances, CES indices, aggregate consumption, and employment are identical. The economies differ only in how these prices are paired inside firms: in C the cheap seller is cheap for both products; in R the identity of the cheap seller changes across products.

Define

$$H = \frac{\varepsilon}{4}(r + \delta)^2, \quad L = \frac{\varepsilon}{4}(r - \delta)^2, \quad \tau = \frac{c}{2}. \quad (22)$$

Proposition 1 (The Organization of Price Gaps Can Matter for Nominal Transmission).

Suppose the initial states satisfy $\varepsilon r^2/4 < \tau$ and, after the nominal expenditure shock,

$$L < \tau < H, \quad H - \tau < K < 2(H - \tau). \quad (23)$$

Then, in the local quadratic model, only the initially cheap type in C adjusts, and it reprices both products; the other type in C and both types in R remain inactive. The two economies have identical product-specific initial price distributions but different impact responses. To first order,

$$\Delta \log P_C \simeq (r + \delta)/2, \quad \Delta \log P_R = 0, \quad \Delta \log C_C \simeq (\delta - r)/2, \quad \Delta \log C_R = \delta. \quad (24)$$

The logic is transparent. For the cheap C type, both products generate the high gross gain H , so their combined net surplus $2(H - \tau)$ covers K . For each R type, only one product generates H while the other yields $L < \tau$; the single profitable product generates only $H - \tau < K$, so the common cost is not activated. The expensive C type has two

low-gain products and also remains inactive. Appendix C provides the proof.

When does direction matter? Proposition 1 is the intermediate case of a simple three-regime comparison in the local quadratic model. Maintaining the same assumptions on the initial state and $L < \tau < H$, let $S_H \equiv H - \tau > 0$ denote the net product-level surplus associated with a large gap:

1. If $K < S_H$, the cheap type in C reprices both products while each type in R reprices its single large-gap product. Each product therefore has the same post-shock price distribution in C and R, so the price-index and consumption responses coincide to first order. Repricing labor need not coincide because more firms pay the common cost in R.
2. If $S_H < K < 2S_H$, Proposition 1 applies: two large gaps in the same firm activate repricing, while one large gap alone does not, so the organization of gaps changes the aggregate response.
3. If $K > 2S_H$, neither economy reprices.

These are first-order regime statements, not exact CES cutoff values. For the parameters used below, $S_H = 0.00304$ and $2S_H = 0.00608$, so $K = .0045$ lies inside the intermediate local window. General-equilibrium feedback through product price indices can shift the exact boundaries; Appendix C verifies directly that the reported action profile is strict at the numerical parameterization. The economic condition is therefore precise: the direction of gaps matters when the common cost is large enough to stop a seller with one large gap but not a seller with two large gaps.

Figure 6 shows the local geometry. The inaction region is no longer a circle: the individual repricing cost creates kinks because small product-level gaps are not worth correcting even after the common cost is paid.

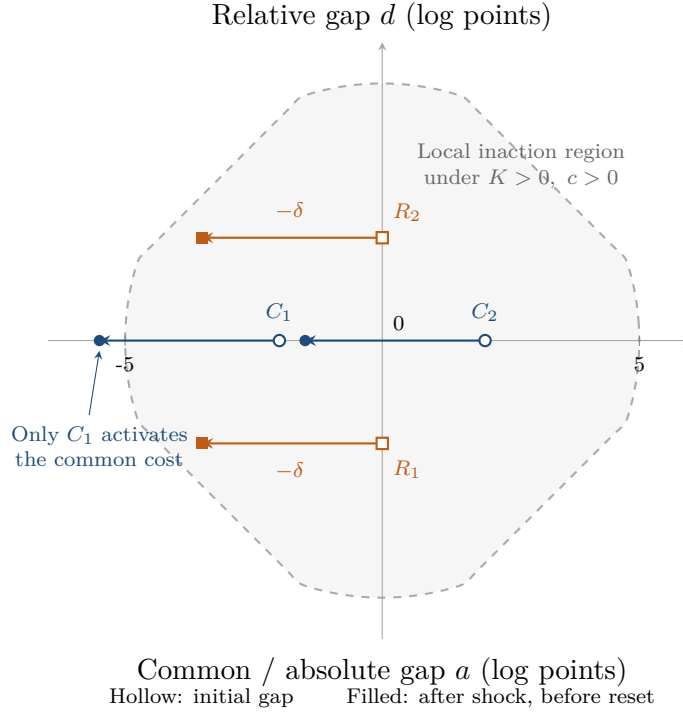


Figure 6: The same nominal expenditure shock shifts differently organized gaps
Notes: Dashed line: local inaction set from equation (20). Parameters are $\varepsilon = 6$, $K = .0045$, $c = .003$, $r = .02$, and $\delta = .035$; coordinates are multiplied by 100. Hollow markers are initial gaps and filled markers post-shock gaps before repricing. Blue circles denote C and orange squares R.

Table 7 establishes the same selection pattern in an exact CES equilibrium. The same nominal expenditure shock produces a larger price response and a smaller consumption response in C. Initial price distributions and the shock are held fixed; inflation and adjustment frequencies are outcomes.

Table 7: Identical initial product-price distributions, different impact responses

	Common gaps	Relative gaps
Nominal expenditure (log points)	3.500	3.500
Price index (log points)	2.836	0.000
Real consumption (log points)	0.664	3.500
Employment (percent)	0.938	3.562
Firms with any reset (percent)	50.000	0.000

Notes: Exact temporary equilibria with $\varepsilon = 6$, $\chi = 5/6$, $K = .0045$, $c = .003$, $r = .02$, and $\Delta \log M = .035$; each type has mass 1/2. Log points are 100 times the log change and employment is a percentage change. Initial product-price distributions are identical across economies. Parameters are illustrative; Appendix C verifies the equilibrium.

The first-order expressions in Proposition 1 imply a 2.75-log-point price response and a

0.75-log-point consumption response in C. Table 7 gives 2.836 and 0.664 log points in the exact CES equilibrium. The small differences reflect curvature and general-equilibrium feedback through the product price indices; the selection pattern is unchanged.

The within-seller adjustment moments indicate partial but limited synchronization: both products adjust in 8.67 percent of observed pairs, compared with 7.01 percent under independence at the same marginal adjustment frequency. These descriptive moments do not identify the model’s cost parameters or desired-price gaps; Online Appendix Section A.14 reports the full set of pair moments and their construction.

6.3 Relative Shocks Can Activate Full Nominal Pass-Through

Relative-price adjustment does not inherently imply aggregate rigidity. If a relative shock makes both product-level gaps sufficiently large, all firms can reprice both products and a nominal expenditure shock can pass through fully. R exhibits no pass-through in the first experiment because its product-level surpluses do not cover the common cost, not because relative pricing is intrinsically rigid.

Start with inherited prices equal to one and set $A_{gi} = \exp(-z_{gi})$, with $\mathbf{z}_1 = (-s, s)$ and $\mathbf{z}_2 = (s, -s)$. Reset prices are $p_{gi}^* = M \exp(z_{gi})$. Compare the same productivities and inherited prices at $M = 1$ and $M = e^\delta$. For $s = .08$ and $\delta = .035$, all firms strictly prefer to reprice both products in both counterfactuals; Appendix C.5 verifies their incentives using exact profits. Consequently,

$$\frac{p_{gi}(e^\delta)}{p_{gi}(1)} = \frac{P(e^\delta)}{P(1)} = e^\delta, \quad \Delta \log C = \delta - \Delta \log P = 0. \quad (25)$$

Product demands and employment, including repricing labor, are unchanged. Impact neutrality is exact conditional on universal adjustment in both counterfactuals, each of which inherits the same prices and independently chooses its adjustment subset. The two experiments therefore imply no universal ranking: selection depends on the configuration of gaps relative to the individual and common adjustment thresholds.

Separately, at $M = 1$, relative repricing raises each product’s log-price standard deviation from zero to s without changing its mean log price across sellers. This aggregation result is distinct from conditional neutrality.²

6.4 Relation to the Empirical Evidence and the Literature

The adjustment technology follows the partial-synchronization logic of Bonomo, Carvalho, Kryvtsov, Ribon, and Rigato (2023): a common fixed cost creates economies of scope, while a product-level cost makes the repriced subset endogenous. The polar cases $c = 0$ and $K = 0$ connect to full-vector adjustment and independent product decisions, respectively (Midrigan, 2011; Alvarez and Lippi, 2014). Holding each product’s marginal price distribution fixed, our illustration isolates the within-firm organization of gaps as an additional state variable; it matters when the common cost makes joint versus individual activation consequential. This complements the broader point that monetary effects depend on the distribution of price gaps rather than adjustment frequency alone (Álvarez, Le Bihan, and Lippi, 2016).

Search and seller–product heterogeneity, strategic interaction under staggered adjustment, and search frictions provide complementary mechanisms linking observed price dispersion and monetary non-neutrality (Kaplan, Menzio, Rudanko, and Trachter, 2019; Wang and Werning, 2022; Head, Liu, Menzio, and Wright, 2012). Evidence across inflation regimes and price concepts adds further margins (Álvarez, Beraja, González-Rozada, and Neumeyer, 2019; Sheremirov, 2020). Our narrower result is that the amount of observed dispersion alone does not determine nominal transmission: its relation to desired prices and its organization within multiproduct firms can also matter.

²The arithmetic mean and CES index change. Appendix C.6 derives both. A stable empirical mean of deflated prices should not be interpreted as a constant model price index.

7 Conclusion

The long-run widening of retail price dispersion in Montevideo primarily reflects a growing separation of sellers' relative positions across products, rather than a widening gap between uniformly cheap and uniformly expensive sellers. Across economic sellers, same-good dispersion rises by 2.29 percentage points between the initial and final twelve-month averages, despite a partial reversal after 2021–2022. The widening is present among retail chains, and seller-product positions account for most of the increase in across-seller variance, with both category-wide and within-category positions contributing. Average real prices do not follow a comparable upward path, and the contemporaneous inflation coefficient remains small and statistically insignificant when inflation is measured at the category level. The composition of dispersion is comparable to that documented for the United States; what the long panel adds is evidence on which component widens over time.

The price-history evidence helps interpret this change. On spells with observed beginnings, most of the fitted seller-product variance increase remains after reversing CPI erosion since the last nominal adjustment. This is a reconstruction from historical reset values on current comparison sets, rather than a distribution of prices newly chosen at a common date. Its allocation is conditional on a support whose trend is approximately 61 percent of the full-sample trend. Together with the distinct movements in adjustment frequency, size, and observed price age, the result shows why a single timing statistic does not summarize the evolution of price setting.

The model illustrates a possible monetary implication. With a common repricing cost and an additional product-level adjustment cost, economies with identical initial price distributions for every good can respond differently to the same nominal expenditure shock when the common cost is strong enough for the combination of gaps to affect whether repricing is activated. When relative shocks already induce universal adjustment, full nominal pass-through is instead possible. The empirical decomposition measures observed

positions rather than desired-price gaps, so it does not quantify these monetary responses. It supplies moments on the long-run composition of dispersion and its relationship to price histories. The illustration shows why this composition deserves attention: alongside the amount of same-good dispersion, the organization of price gaps within multiproduct firms can shape adjustment decisions and aggregate price and consumption responses.

References

- ADAMS, B., AND K. R. WILLIAMS (2019): “Zone Pricing in Retail Oligopoly,” *American Economic Journal: Microeconomics*, 11(1), 124–156.
- ÁLVAREZ, F., M. BERAJA, M. GONZÁLEZ-ROZADA, AND P. A. NEUMEYER (2019): “From Hyperinflation to Stable Prices: Argentina’s Evidence on Menu Cost Models,” *The Quarterly Journal of Economics*, 134(1), 451–505.
- ÁLVAREZ, F., H. LE BIHAN, AND F. LIPPI (2016): “The Real Effects of Monetary Shocks in Sticky Price Models: A Sufficient Statistic Approach,” *American Economic Review*, 106(10), 2817–2851.
- ALVAREZ, F., AND F. LIPPI (2014): “Price Setting With Menu Cost for Multiproduct Firms,” *Econometrica*, 82(1), 89–135.
- ATKIN, D., AND D. DONALDSON (2015): “Who’s Getting Globalized? The Size and Implications of Intra-national Trade Costs,” Working Paper 21439, National Bureau of Economic Research.
- BILS, M., P. J. KLENOW, AND B. A. MALIN (2012): “Reset Price Inflation and the Impact of Monetary Policy Shocks,” *American Economic Review*, 102(6), 2798–2825.
- BONOMO, M., C. CARVALHO, O. KRYVTSOV, S. RIBON, AND R. RIGATO (2023): “Multi-Product Pricing: Theory and Evidence from Large Retailers,” *The Economic Journal*, 133(651), 905–927.
- BORRAZ, F., A. CAVALLO, R. RIGOBON, AND L. ZIPITRÍA (2016): “Distance and Political Boundaries: Estimating Border Effects under Inequality Constraints,” *International Journal of Finance & Economics*, 21(1), 3–35.
- BORRAZ, F., AND J. SALDAIN (2017): “Variance Decomposition of Prices in an Emerging Economy,” *Monetaria*, (2), 243–269.

- BORRAZ, F., AND L. ZIPITRÍA (2012): “Retail Price Setting in Uruguay,” *Economia*, 12(2), 77–109.
- (2022): “Varieties as a Source of Law of One Price Deviations,” *International Economics*, 172, 1–14.
- CAGLAYAN, M., A. FILIZTEKIN, AND M. T. RAUH (2008): “Inflation, Price Dispersion, and Market Structure,” *European Economic Review*, 52(7), 1187–1208.
- CAMPBELL, J. R., AND B. EDEN (2014): “Rigid Prices: Evidence from U.S. Scanner Data,” *International Economic Review*, 55(2), 423–442.
- DARUICH, D., AND J. KOZLOWSKI (2023): “Macroeconomic Implications of Uniform Pricing,” *American Economic Journal: Macroeconomics*, 15(3), 64–108.
- DELLAVIGNA, S., AND M. GENTZKOW (2019): “Uniform Pricing in U.S. Retail Chains,” *The Quarterly Journal of Economics*, 134(4), 2011–2084.
- DVIR, E., AND G. STRASSER (2018): “Does Marketing Widen Borders? Cross-Country Price Dispersion in the European Car Market,” *Journal of International Economics*, 112, 134–149.
- EICHENBAUM, M., N. JAIMOVICH, AND S. REBELO (2011): “Reference Prices, Costs, and Nominal Rigidities,” *American Economic Review*, 101(1), 234–62.
- GLANDON, P. J. (2018): “Sales and the (Mis)Measurement of Price Level Fluctuations,” *Journal of Macroeconomics*, 58, 60–77.
- HEAD, A., L. Q. LIU, G. MENZIO, AND R. WRIGHT (2012): “Sticky Prices: A New Monetarist Approach,” *Journal of the European Economic Association*, 10(5), 939–973.
- HITSCH, G. J., A. HORTAÇSU, AND X. LIN (2021): “Prices and promotions in U.S. retail markets,” *Quantitative Marketing and Economics (QME)*, 19(3), 289–368.

- INSTITUTO NACIONAL DE ESTADÍSTICA (2011): “Índice de Precios del Consumo: Cambio de base – Diciembre 2010. Nota metodológica,” Discussion paper, Instituto Nacional de Estadística, Montevideo.
- (2022): “Metodología Índice de Precios del Consumo,” Discussion paper, Instituto Nacional de Estadística, Montevideo, Base octubre 2022 = 100.
- KAPLAN, G., AND G. MENZIO (2015): “The Morphology Of Price Dispersion,” *International Economic Review*, 56(4), 1165–1206.
- KAPLAN, G., G. MENZIO, L. RUDANKO, AND N. TRACHTER (2019): “Relative Price Dispersion: Evidence and Theory,” *American Economic Journal: Microeconomics*, 11(3), 68–124.
- KLACZKO, I. (2025): “Price Dispersion in Uruguay,” *Latin American Economic Review*, 35, 1–29.
- KLENOW, P. J., AND O. KRYVTSOV (2008): “State-Dependent or Time-Dependent Pricing: Does It Matter for Recent U.S. Inflation?,” *The Quarterly Journal of Economics*, 123(3), 863–904.
- KLENOW, P. J., AND B. A. MALIN (2011): “Microeconomic Evidence on Price-Setting,” in *Handbook of Monetary Economics*, ed. by B. M. Friedman, and M. Woodford, vol. 3A, pp. 231–284. Elsevier.
- MIDRIGAN, V. (2011): “Menu Costs, Multiproduct Firms, and Aggregate Fluctuations,” *Econometrica*, 79(4), 1139–1180.
- NAKAMURA, A., E. NAKAMURA, AND L. NAKAMURA (2011): “Price dynamics, retail chains and inflation measurement,” *Journal of Econometrics*, 161, 47–55.
- NAKAMURA, E., AND J. STEINSSON (2008): “Five Facts about Prices: A Reevaluation of Menu Cost Models,” *Quarterly Journal of Economics*, 123(4), 1415–1464.

- REINSdorf, M. (1994): “New Evidence on the Relation Between Inflation and Price Dispersion,” *American Economic Review*, 84(3), 720–731.
- SARA-ZAROR, F. (2025): “Inflation, Price Dispersion, and Welfare: The Role of Consumer Search,” Manuscript, October 30.
- SHEREMIROV, V. (2020): “Price dispersion and inflation: New facts and theoretical implications,” *Journal of Monetary Economics*, 114(C), 59–70.
- URUGUAY XXI (2018): “Oportunidades de inversión. SECTOR RETAIL,” Discussion paper, Uruguay XXI, Uruguay, Available at: <https://www.uruguayxxi.gub.uy/uploads/informacion/26f81e9ae50b0543403b87b17fa8429babbee409.pdf>.
- WANG, O., AND I. WERNING (2022): “Dynamic Oligopoly and Price Stickiness,” *American Economic Review*, 112(8), 2815–2849.

Online Appendix A. Additional Evidence and Robustness

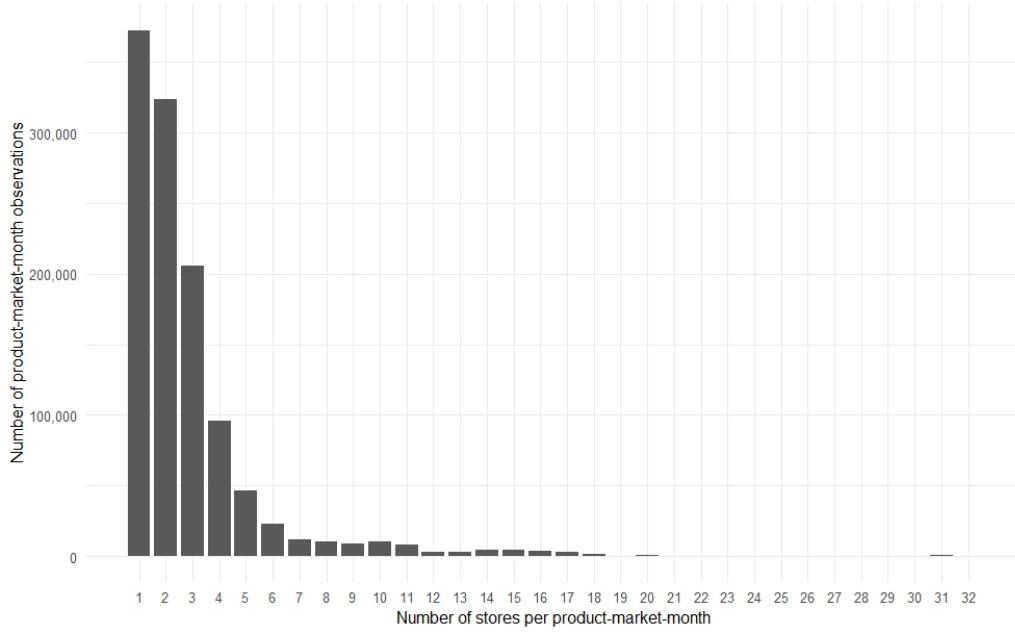
Appendix A documents data coverage, flexible time paths, inflation comparisons, composition and retail-chain robustness, the decomposition of seller-product dispersion, and price-history and adjustment checks, including the moments linked to the model. Appendix B separately documents the disaggregated Montevideo CPI construction and category matching used in the inflation comparison. Unless otherwise noted, appendix regressions retain the controls, weights, and inference of the corresponding main-text specification; table notes report sample restrictions, constructions, or departures from those conventions. Component shares follow the fitted-trend convention defined in Section 4.1. Throughout Appendix A, ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Data Coverage and the Long-Run Dispersion Result

A.1 Sample Coverage

Price dispersion is defined only when multiple stores report the same product in the same market and month. Figure A.1 therefore reports the distribution of the number of stores underlying product–market–month comparisons. This is the relevant comparison size for the baseline dispersion measure; sensitivity to small cells is examined separately in Section A.6.

Figure A.1: Distribution of Stores per Product–Market–Month.



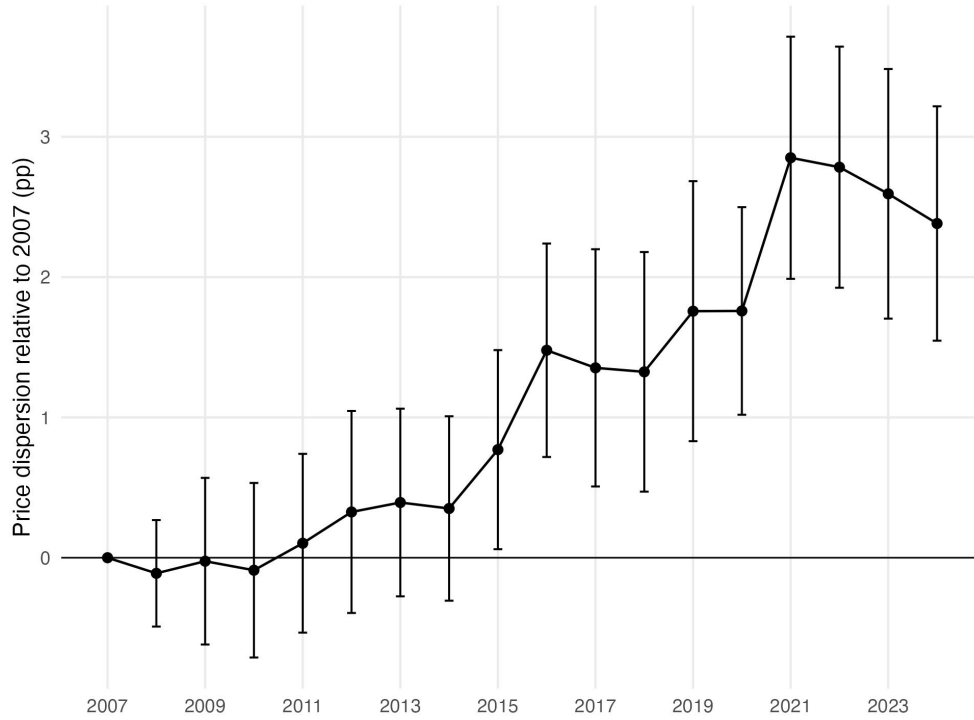
Notes: The figure shows the distribution of the number of stores reporting the same product in the same market and month. This is the number of observations available for constructing product–market–month price dispersion.

A.2 Flexible Time Path and Average Real Prices

The baseline estimates summarize the long-run evolution of price dispersion using a linear monthly trend. Figure A.2 relaxes this restriction and replaces the linear trend with year indicators, while controlling for product-by-market and calendar-month fixed effects. The figure confirms that the long-run increase is not an artifact of imposing linearity. Dispersion rises particularly strongly from the mid-2010s, reaches high levels around 2021–2022, and remains substantially above its 2007 level at the end of the sample.

For a direct comparison on the support where price dispersion is defined, Figures A.3 and A.4 restrict attention to product–market–month cells with at least two stores. Figure A.3 estimates unrestricted monthly time effects after absorbing product-by-market fixed effects. To avoid making a single month the benchmark, the series is normalized so that the average effect over the first 12 months, April 2007 through March 2008, equals zero. The thin line reports the monthly effects, the thick line a 12-month moving average, and the dashed line a fitted linear trend.

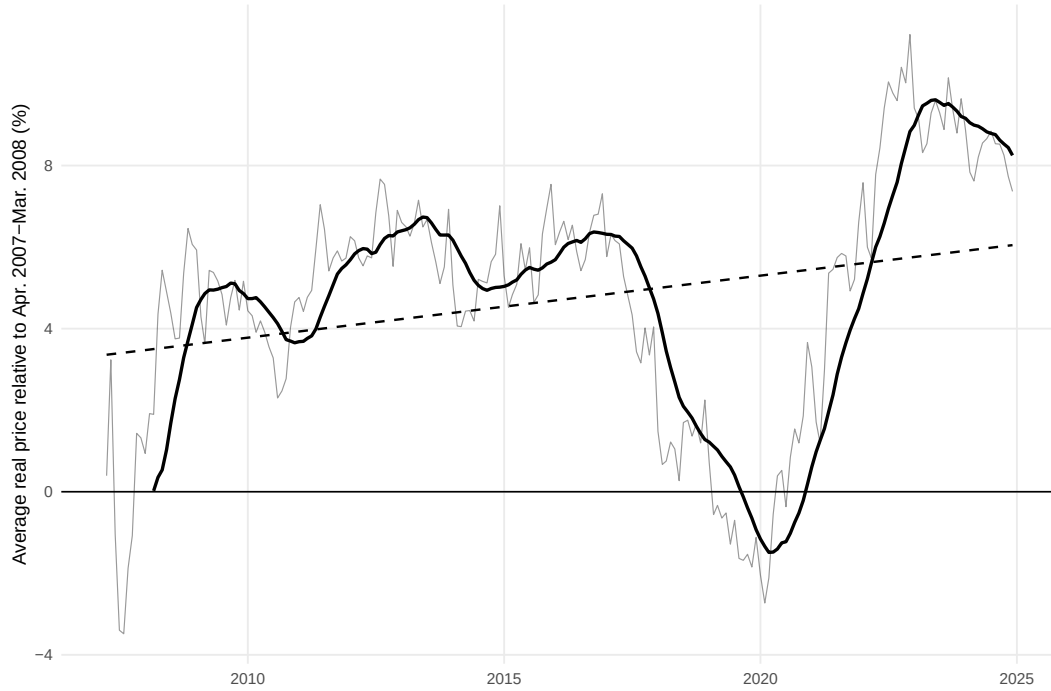
Figure A.2: Price Dispersion: Flexible Year Effects.



Notes: Year effects from the baseline dispersion specification with product-by-market and calendar-month fixed effects. Estimates are store-count weighted; bars are 95 percent confidence intervals from three-way clustered standard errors.

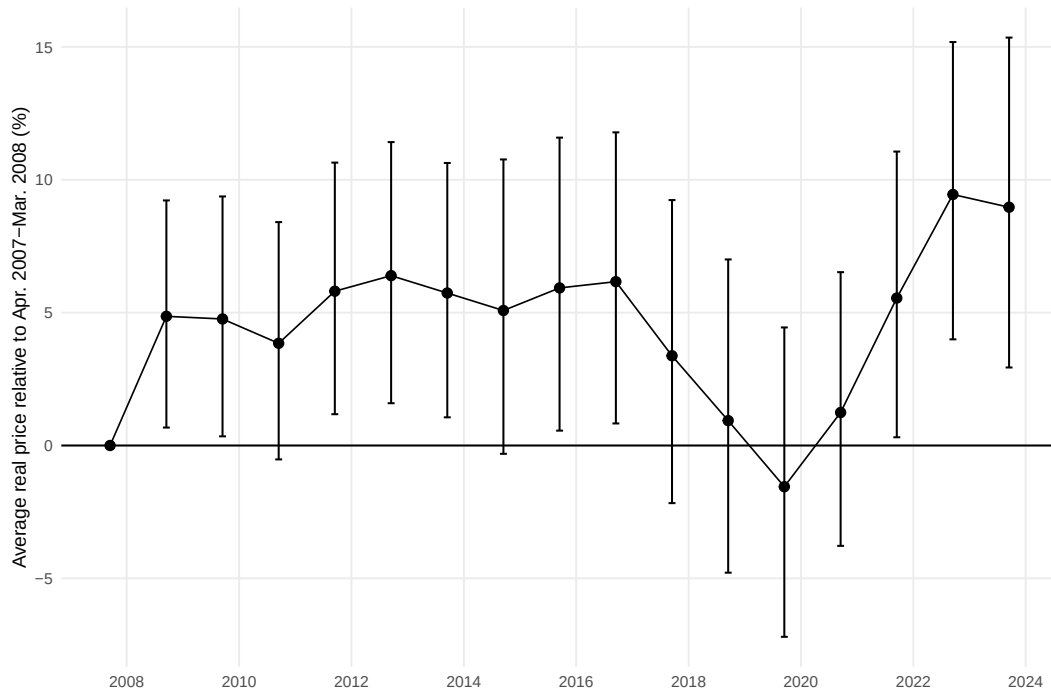
Average real prices display substantial medium-run movements rather than a path resembling the secular widening of dispersion. They rise early in the sample, remain in a relatively narrow range for much of 2009–2017, fall sharply around 2018–2020, and subsequently recover. Figure A.4 verifies that the early increase is not an artifact of using the incomplete 2007 calendar year as the reference category. The figure divides the sample into complete April–March 12-month blocks and expresses each block relative to April 2007–March 2008.

Figure A.3: Average Real Prices: Monthly Time Effects.



Notes: Monthly time effects for average log CPI-adjusted price on cells with at least two stores and product-by-market fixed effects. Effects are normalized to average zero over April 2007–March 2008. Thick line: 12-month moving average; dashed line: linear fit.

Figure A.4: Average Real Prices: Non-Overlapping 12-Month Blocks.



Notes: Non-overlapping April–March block effects for average log CPI-adjusted price on cells with at least two stores, with product-by-market and calendar-month fixed effects. April 2007–March 2008 is omitted; bars are 95 percent confidence intervals from three-way clustered standard errors.

A.3 Average-Price, Inflation, and Pandemic-Period Robustness

A first specification concern is that average price and price dispersion are constructed from the same underlying product–market–month prices. Conditioning on average log real price could therefore affect the estimated time trend. Table A.1 re-estimates the preferred specifications without this control, while leaving the sample, fixed effects, weights, and inference unchanged.

Table A.1: Long-Run Price-Dispersion Trend With and Without Average Price Control.

Outcome	With average price		Without average price	
	Time	Std. Error	Time	Std. Error
Across sellers	0.0192***	0.0023	0.0184***	0.0021
Aggregate	0.0166***	0.0020	0.0159***	0.0019

Notes: Both columns use the same sample and baseline specification; the only difference is whether average log CPI-adjusted price is included as a control.

Removing average log real price has little effect on the result. The estimated trend declines from 0.0166 to 0.0159 for aggregate dispersion and from 0.0192 to 0.0184 for across-seller dispersion, changes of approximately 4 percent. Both trends remain positive and highly statistically significant. The long-run increase in dispersion is therefore not generated by conditioning on the average price of the same product–market–month observations.

A related macroeconomic concern is the relationship between inflation and price dispersion. This is particularly relevant because Sheremirov (2020) documents positive comovement between inflation and regular-price dispersion, in contrast to the negative comovement he finds for posted-price dispersion. Because our monthly modal price measure attenuates temporary promotional price changes, it is useful to examine whether inflation dynamics can account for the long-run widening documented in our data.

The main text reports the aggregate CPI comparison, while the category-specific inflation exercise uses the matched official Montevideo CPI series described below.

Table [A.2](#) extends this comparison beyond contemporaneous inflation. A potential concern is that inflation may affect price dispersion with a lag or through accumulated inflationary pressure, so controlling only for current inflation may fail to capture the relevant dynamics. We therefore consider current 12-month inflation, inflation over the previous 12 months, current and previous inflation jointly, cumulative 24-month inflation, and the trailing 12-month volatility of monthly inflation. General CPI is our primary macroeconomic measure, while food-price inflation in Montevideo provides a sector-specific robustness check.

The estimated long-run dispersion trend is very stable across these specifications. For across-seller dispersion, adding the alternative general CPI controls changes the estimated time coefficient by less than 3 percent relative to the corresponding same-sample baseline. For aggregate dispersion, the largest change is approximately 3.4 percent. Thus, none of the inflation controls materially reduces the estimated long-run trend.

The table reports the sensitivity of the time coefficient, rather than the coefficients on inflation themselves. Its result is the stability of the fitted dispersion trend under the controls considered, not a test of independence or an estimate of the causal effect of inflation.

Finally, Table [A.3](#) examines whether the exceptional observations around the pandemic drive the estimated trend. Excluding 2020 leaves the aggregate and across-seller trends virtually unchanged. Excluding both 2020 and 2021 reduces the coefficients modestly, but the estimated long-run increases remain positive and highly statistically significant.

Table A.2: Inflation Dynamics and the Long-Run Price-Dispersion Trend.

Inflation control	Baseline Time	Controlled Time	Change in Time (%)	Observations
<i>Aggregate dispersion</i>				
General CPI: current 12m	0.0166***	0.0169***	1.7	770,942
General CPI: previous 12m	0.0166***	0.0165***	-0.7	770,942
General CPI: current + previous 12m	0.0166***	0.0168***	0.8	770,942
General CPI: cumulative 24m	0.0166***	0.0170***	2.2	770,942
General CPI: 12m volatility	0.0166***	0.0168***	0.9	770,942
General CPI: current 12m + volatility	0.0166***	0.0172***	3.4	770,942
Food CPI: current 12m	0.0187***	0.0187***	0.2	616,239
Food CPI: previous 12m	0.0191***	0.0190***	-0.2	566,103
Food CPI: current + previous 12m	0.0191***	0.0193***	1.1	566,103
Food CPI: cumulative 24m	0.0191***	0.0195***	2.1	566,103
Food CPI: 12m volatility	0.0187***	0.0189***	1.1	616,239
<i>Across sellers dispersion</i>				
General CPI: current 12m	0.0192***	0.0196***	2.2	648,318
General CPI: previous 12m	0.0192***	0.0190***	-0.7	648,318
General CPI: current + previous 12m	0.0192***	0.0194***	1.4	648,318
General CPI: cumulative 24m	0.0192***	0.0196***	2.1	648,318
General CPI: 12m volatility	0.0192***	0.0192***	0.2	648,318
General CPI: current 12m + volatility	0.0192***	0.0197***	2.9	648,318
Food CPI: current 12m	0.0216***	0.0216***	0.4	516,234
Food CPI: previous 12m	0.0216***	0.0216***	-0.0	473,314
Food CPI: current + previous 12m	0.0216***	0.0218***	0.9	473,314
Food CPI: cumulative 24m	0.0216***	0.0219***	1.3	473,314
Food CPI: 12m volatility	0.0216***	0.0221***	2.4	516,234

Notes: Current 12m is year-on-year inflation; previous 12m covers $t - 24$ to $t - 12$; cumulative 24m covers $t - 24$ to t ; volatility is the trailing 12-month standard deviation of monthly log inflation. General CPI is the primary measure and Montevideo food CPI a sector-specific check. Baseline and controlled estimates use identical observations within each row. Change in Time is the percentage change in the estimated trend after adding the indicated control.

Table A.3: Long-Run Price-Dispersion Trend Excluding the Pandemic Period.

	Full sample	Exclude 2020	Exclude 2020–2021
Aggregate	0.0166*** (0.0020)	0.0167*** (0.0020)	0.0156*** (0.0019)
Observations	770,942	726,580	680,894
Across sellers	0.0192*** (0.0023)	0.0192*** (0.0023)	0.0181*** (0.0021)
Observations	648,318	612,062	573,514

Notes: Each entry is the monthly time-trend coefficient under the indicated calendar exclusion; otherwise the baseline specifications are unchanged.

A.4 Price Dispersion and Category-Specific Inflation

This exercise asks whether using aggregate inflation conceals a contemporaneous association between inflation and price dispersion. We match dispersion for each product–market–month to official inflation for the corresponding consumption category in Montevideo. It refines the aggregate comparison in Section 3.2 by allowing inflation to differ across categories.

Following the regression approach of Sheremirov (2020), our specification with product–market and date fixed effects is

$$\sigma_{imt} = \beta\pi_{c(i)t} + \alpha_{im} + \tau_t + \sum_{\ell=1}^{12} a_{\ell}\Delta\sigma_{im,t-\ell} + \sum_{\ell=1}^{12} b_{\ell}\Delta\pi_{c(i),t-\ell} + \varepsilon_{imt}, \quad (26)$$

where i denotes a product, m a neighborhood market, t a calendar month, and $c(i)$ the category of product i . Dispersion, σ_{imt} , is 100 times the sample standard deviation of log monthly modal prices across physical stores selling the same product in the same market and month. Cells require at least two stores. A common deflator within a cell leaves this dispersion unchanged. Inflation is $\pi_{ct} = 1200(\log I_{ct} - \log I_{c,t-1})$, where I_{ct} is the linked category CPI: it is monthly log inflation expressed at an annualized rate, not twelve-month inflation. Appendix Section B documents the sources, coverage, category correspondence, and linking of the two CPI vintages.

Table A.4 reports OLS estimates for January 2011–December 2024, with equal weight per observed product–market–month. Column (1) includes product and month-of-year fixed effects. Column (2) replaces product effects with product–market effects. Column (3) additionally replaces month-of-year effects with calendar-date effects. Column (4) repeats column (3) on the sample eligible for the lag controls. Column (5) adds twelve monthly lags of the first differences of both dispersion and inflation, as in equation (26). The lag terms serve as controls; the reported coefficient of interest is the contemporaneous β . Missing calendar months are not interpolated. Standard errors are Driscoll–Kraay with twelve lags.

This is an adaptation rather than an exact replication of Sheremirov (2020). Official inflation varies across categories and months but is common to all neighborhoods in Montevideo; with date effects, identification uses category inflation movements net of their common monthly component. We use monthly modal prices and do not apply expenditure weights. Seasonality is controlled by month-of-year or date effects rather than X-12 adjustment; these effects absorb common seasonal patterns, not necessarily category-specific seasonality.

Table A.4: Price dispersion and category inflation

Dependent Variable: Model:	Price dispersion (100 x SD of log prices)				
	(1)	(2)	(3)	(4)	(5)
<i>Variables</i>					
Category inflation (annualized log points)	-0.0024 (0.0033)	-0.0021 (0.0032)	-0.0020 (0.0017)	-0.0019 (0.0019)	-0.0017 (0.0013)
12 lags of changes in dispersion and inflation	No	No	No	No	Yes
<i>Fixed-effects</i>					
Product	Yes				
Month of year	Yes	Yes			
Product-market		Yes	Yes	Yes	Yes
Calendar date			Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	662,703	662,642	662,642	498,633	498,633
R ²	0.10605	0.37778	0.39344	0.43259	0.52707

Driscoll-Kraay (L=12) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dispersion is $100 \times SD(\log p)$ across physical stores within product–market–month; category inflation is the monthly log CPI change times 1200. OLS uses equal cell weights and Driscoll–Kraay standard errors with 12 lags. Columns (4) and (5) use the same eligible sample; column (5) adds 12 lags of each first difference.

The contemporaneous inflation coefficient is negative, small, and statistically insignificant throughout. With product–market and date effects it is -0.0020 (standard error 0.0017); adding the lag controls yields -0.0017 (standard error 0.0013). Restricting the sample before introducing these controls changes the estimate little. Thus, using category-specific inflation does not reveal a statistically significant contemporaneous association hidden by the aggregate measure. This finding complements the evidence on the long-run

widening of dispersion; it does not establish general independence between inflation and dispersion or identify a causal effect of inflation.

A.5 Store, Seller, and Product Composition

This subsection tests whether changes in store coverage or in the monitored product list generate the widening.

Table A.5 recomputes physical-store dispersion using stores observed in 2007 and reports monthly trends under alternative fixed effects. It asks whether widening remains among the original store cohort; it does not require every original store to report in every month. The trends remain positive and statistically significant.

Table A.5: Baseline Price-Dispersion Trend Estimates: Initial Sample Stores.

Dependent Variable: Model:	(1)	(2)	SD (in %)		(5)	(6)
			(3)	(4)		
<i>Variables</i>						
Av. Price	-1.869*** (0.4627)	-8.129*** (1.536)	-7.671*** (1.429)	-7.666*** (1.432)	-6.449*** (1.247)	-6.443*** (1.248)
Time	0.0127*** (0.0024)	0.0125*** (0.0025)	0.0128*** (0.0023)	0.0128*** (0.0023)	0.0126*** (0.0023)	0.0127*** (0.0023)
<i>Fixed-effects</i>						
Market	Yes		Yes	Yes		
Product		Yes	Yes	Yes		
Month				Yes		Yes
Product $\times$ Market					Yes	Yes
<i>Fit statistics</i>						
Observations	623,312	623,312	623,312	623,312	623,222	623,222
R ²	0.14115	0.15049	0.24639	0.24663	0.37479	0.37502
Within R ²	0.04648	0.04462	0.04629	0.04629	0.04242	0.04242

Clustered (Category & Market & Time) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The sample is restricted to stores already observed in 2007. The dependent variable and baseline regression conventions follow Section 3.1; fixed effects are indicated in the table.

Table A.6 excludes markets in the top decile of store entry after the initial year and re-estimates the monthly trend. This checks whether high-entry neighborhoods drive the

result; the coefficient remains similar to the baseline. It is a sample restriction, not an estimate of the causal effect of entry.

Table A.6: Baseline Price-Dispersion Trend Estimates: Excluding High-Entry Markets.

Dependent Variable: Model:	(1)	(2)	SD (in %)		(5)	(6)
			(3)	(4)		
<i>Variables</i>						
Av. Price	-2.186*** (0.5135)	-8.448*** (1.659)	-7.731*** (1.556)	-7.725*** (1.558)	-6.649*** (1.345)	-6.641*** (1.347)
Time	0.0172*** (0.0023)	0.0160*** (0.0026)	0.0168*** (0.0023)	0.0168*** (0.0023)	0.0167*** (0.0022)	0.0167*** (0.0022)
<i>Fixed-effects</i>						
Market	Yes		Yes	Yes		
Product		Yes	Yes	Yes		
Month				Yes		Yes
Product × Market					Yes	Yes
<i>Fit statistics</i>						
Observations	617,916	617,916	617,916	617,916	617,859	617,859
R ²	0.15517	0.16490	0.26171	0.26187	0.39532	0.39547
Within R ²	0.06192	0.05077	0.05257	0.05256	0.05134	0.05132

Clustered (Category & Market & Time) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The sample excludes markets in the top decile of store entry after the initial sample year; stores first observed in 2007 are part of the initial stock. The dependent variable and baseline regression conventions follow Section 3.1.

Product composition raises a related but distinct concern. The monitored product list underwent a major revision in November 2010, so changes in the set of products followed over time could in principle generate an apparent increase in dispersion. A simple post-2011 restriction addresses the discrete revision, but does not by itself hold the product set fixed. We therefore impose progressively stronger restrictions on product composition.

The first restrictions hold product support fixed over the long run. Restricting the sample to the 70 products observed during both the first and final 12 months produces an across-seller trend of 0.0179, compared with 0.0192 in the full sample. Requiring individual product–market pairs to be observed near both endpoints is more demanding

Table A.7: Product-Cohort and Fixed-Product Robustness.

Sample	Time	Std. Error	Products	Observations
<i>Aggregate dispersion</i>				
Full sample	0.0166***	0.0020	125	770,942
Products observed in first and final 12m	0.0158***	0.0020	70	499,010
Product-market pairs observed in first and final 12m	0.0155***	0.0020	68	381,894
2011 onward: all products	0.0184***	0.0026	125	662,642
2012 onward: all products	0.0187***	0.0030	125	612,044
2012 onward: pre-revision products	0.0186***	0.0030	73	376,919
2012 onward: 2010–11 entrants	0.0179***	0.0043	50	227,859
<i>Across sellers dispersion</i>				
Full sample	0.0192***	0.0023	125	648,318
Products observed in first and final 12m	0.0179***	0.0024	70	423,246
Product-market pairs observed in first and final 12m	0.0171***	0.0022	67	321,970
2011 onward: all products	0.0214***	0.0029	125	555,836
2012 onward: all products	0.0216***	0.0034	125	512,652
2012 onward: pre-revision products	0.0207***	0.0035	73	317,967
2012 onward: 2010–11 entrants	0.0221***	0.0047	50	188,939

Notes: The fixed-product sample retains products observed in both the first and final 12 months; the fixed product–market sample additionally requires endpoint presence of each product–market pair. The 2011-onward row starts after the late-2010 basket revision. The final three rows use the common 2012–2024 window and compare all products with pre-revision products and 2010–2011 entrants. A linear product-age control is absorbed by product fixed effects because age equals calendar time minus a product-specific entry date.

and yields a trend of 0.0171. The corresponding aggregate-dispersion trends are 0.0158 and 0.0155, respectively. All remain positive and highly statistically significant.

The table also separates the effect of the 2010 revision from differences between product cohorts. Restricting the sample to 2011 onward yields an across-seller trend of 0.0214. More importantly, over the common 2012–2024 period, products already present before the revision exhibit a trend of 0.0207, while the 50 products introduced with the November 2010 revision exhibit a trend of 0.0221. For aggregate dispersion, the corresponding estimates are 0.0186 and 0.0179. Thus, both the pre-existing products and the revision cohort display substantial subsequent widening.

The comparisons show widening within long-lived product sets and both entry cohorts. They address observed basket turnover; they do not rule out changes in demand or other conditions affecting products as they age.

Economic-seller composition. Physical-store entry and economic-seller turnover are distinct margins because a new branch can belong to an existing chain. Table A.8 retains stable chain identities, stable chain–market pairs, or seller–market pairs observed near both endpoints. The coefficients remain positive and significant and fall only modestly; the restrictions select long-lived sellers without requiring a fully balanced panel.

Table A.8: Across-Seller Trend with Fixed Seller Composition.

Sample restriction	Product FE	Product + Market FE	Product $\times$ Market FE	Obs.
Baseline	0.0193*** (0.0025)	0.0193*** (0.0024)	0.0192*** (0.0023)	648,318
Reconstructed	0.0193*** (0.0025)	0.0193*** (0.0024)	0.0192*** (0.0023)	648,318
Stable chains	0.0183*** (0.0025)	0.0177*** (0.0025)	0.0176*** (0.0024)	589,684
Stable chain–market pairs	0.0177*** (0.0026)	0.0177*** (0.0025)	0.0175*** (0.0024)	532,284
Continuing seller–market pairs	0.0183*** (0.0026)	0.0177*** (0.0025)	0.0174*** (0.0025)	461,807

Notes: *Baseline* uses the canonical across-seller sample and *Reconstructed* rebuilds it from store-level prices. *Stable chains* retains all independents but only chains present near both endpoints; *Stable chain–market pairs* also requires presence in the same market; *Continuing seller–market pairs* applies the endpoint restriction to every economic seller. Endpoint presence uses the first and last 12 months.

A.6 Small-Cell Robustness and Time-Series Inference

Table A.9 addresses two different issues. Panel A checks whether changes in the number of sellers per comparison account for the trend, controlling for seller counts and using fixed-count subsamples. With exactly two sellers, the monthly trend is 0.0185 rather than the baseline 0.0192; changing comparison size cannot explain that restricted-sample increase.

Panel B checks serial dependence in the monthly series. It regresses 213 weighted monthly averages on a trend and calendar-month indicators and reports conventional and Newey–West standard errors with 12 and 24 lags. Positive trends remain significant under

both HAC choices. This tests inference for the stated trend specification, not whether the series is stationary or whether the trend will persist beyond the sample.

Table A.9: Small-Cell Robustness and Time-Series Inference.

Specification	Time	Std. Error	Observations
<i>Panel A. Number of economic sellers underlying across-seller dispersion</i>			
Baseline	0.0192***	0.0023	648,318
Seller-count fixed effects	0.0179***	0.0023	648,318
Exactly 2 sellers	0.0185***	0.0030	323,579
Exactly 3 sellers	0.0164***	0.0027	170,650
At least 3 sellers	0.0182***	0.0022	324,558
<i>Panel B. Monthly aggregate series and time-series inference</i>			
Aggregate — Conventional	0.0160***	0.0005	213
Aggregate — Newey–West (12)	0.0160***	0.0015	213
Aggregate — Newey–West (24)	0.0160***	0.0017	213
Across sellers — Conventional	0.0188***	0.0006	213
Across sellers — Newey–West (12)	0.0188***	0.0020	213
Across sellers — Newey–West (24)	0.0188***	0.0024	213

Notes: Panel A reports the across-seller trend under seller-count controls or fixed-count subsamples. Panel B uses 213 seller-count-weighted monthly averages and reports conventional and Newey–West standard errors with 12 or 24 lags; the point estimate is common across inference methods.

A.7 Retail-Chain Comparison Support and Influence

This check asks whether comparing chain–chain and chain–independent trends only where both are observed selects an unusual sample. Table A.10 reports monthly trends on each pair type’s full support and on their common support. Chain–chain comparisons cover about 77 percent of eligible across-seller cells and chain–independent comparisons about 45 percent; the overlap contains 169,769 cells before fixed-effect singleton removal. Similar full-support trends show that the result does not depend on using the overlap alone.

Table A.10: Price-Dispersion Trends by Seller Pair Type and Sample Support.

Sample	Comparison	Time	Std. Error	Observations
Current paper	Across sellers	0.0192***	0.0023	648,318
All available	Chain–Chain	0.0201***	0.0026	497,295
All available	Chain–Independent	0.0197***	0.0025	292,753
Common support	Chain–Chain	0.0202***	0.0027	169,703
Common support	Chain–Independent	0.0194***	0.0027	169,703
Common support	Difference: CI – CC	-0.0008	0.0023	339,406

Notes: The current-paper row reproduces the preferred across-seller trend. All-available rows use each pair type’s full support; common-support rows require both pair types. The difference is estimated in the stacked common-support specification.

Individual-chain influence. We next remove each chain, reconstruct both dispersion measures, and re-estimate the preferred trends to check whether one chain with an unusual pricing path generates the across-seller or chain–chain trend. Removing a chain can eliminate comparisons with too few remaining sellers, so each omission is evaluated against the original baseline on exactly the same surviving product–market–month cells. Table A.11 reports these matched-support comparisons.

Both trends remain positive and significant after every omission. Excluding the largest chain changes the across-seller trend from 0.0185 to 0.0165 (about 11 percent) and the chain–chain trend from 0.0184 to 0.0160 (about 13 percent). Individual chains affect the magnitude, but none alone generates the widening on the comparisons that remain observable.

Table A.11: Leave-One-Chain-Out Robustness.

Omitted chain	Same-support baseline	Leave-one-out trend	Change (%)	Support retained (%)	Observations
<i>Across sellers dispersion</i>					
ta - ta	0.0185	0.0165***	-11.0	79.9	518,008
red expres	0.0192	0.0191***	-0.2	88.9	576,380
devoto	0.0193	0.0196***	1.4	90.4	585,903
disco	0.0190	0.0197***	4.0	90.3	585,309
devoto express	0.0196	0.0198***	1.1	98.1	635,699
tienda inglesa	0.0197	0.0205***	3.8	96.2	623,705
red market	0.0187	0.0188***	0.5	93.7	607,426
super tico	0.0188	0.0187***	-0.2	95.4	618,381
macromercado mayorista	0.0192	0.0192***	0.1	96.0	622,450
el clon	0.0191	0.0187***	-2.1	97.7	633,175
micro macro	0.0190	0.0189***	-0.7	98.8	640,484
frog	0.0191	0.0188***	-1.7	99.3	643,659
kinko	0.0191	0.0174***	-8.9	99.6	645,782
la colonial	0.0190	0.0189***	-0.8	99.7	646,474
planeta	0.0192	0.0193***	0.6	99.6	645,883
multi ahorro	0.0191	0.0191***	-0.3	99.5	645,172
frigo	0.0191	0.0189***	-0.9	99.5	644,866
los jardines	0.0193	0.0195***	1.1	99.2	643,401
geant	0.0192	0.0193***	0.8	99.8	647,308
iberpark	0.0192	0.0197***	3.0	100.0	648,288
supermercados 5 estrellas	0.0192	0.0190***	-1.0	100.0	648,050
el dorado	0.0192	0.0192***	0.0	99.9	647,872
<i>Chain-Chain dispersion</i>					
ta - ta	0.0184	0.0160***	-13.1	70.7	351,500
red expres	0.0201	0.0200***	-0.5	87.0	432,546
devoto	0.0200	0.0203***	1.3	83.6	415,847
disco	0.0199	0.0208***	4.6	82.7	411,311
devoto express	0.0205	0.0209***	1.9	96.9	481,975
tienda inglesa	0.0209	0.0224***	7.2	94.9	472,040
red market	0.0194	0.0195***	0.4	92.4	459,346
super tico	0.0195	0.0196***	0.3	94.3	469,030
macromercado mayorista	0.0202	0.0202***	0.0	92.8	461,509
el clon	0.0200	0.0194***	-2.6	96.0	477,225
micro macro	0.0198	0.0199***	0.2	97.8	486,210
frog	0.0200	0.0194***	-3.0	99.0	492,452
kinko	0.0200	0.0176***	-11.9	99.4	494,547
la colonial	0.0198	0.0197***	-0.8	99.6	495,347
planeta	0.0201	0.0201***	0.3	98.7	491,052
multi ahorro	0.0201	0.0199***	-0.7	98.7	491,054
frigo	0.0200	0.0198***	-0.9	99.3	493,844
los jardines	0.0203	0.0207***	1.7	99.0	492,378
geant	0.0201	0.0202***	0.4	99.7	496,000
iberpark	0.0201	0.0211***	4.8	100.0	497,216
supermercados 5 estrellas	0.0201	0.0198***	-1.2	99.9	497,006
el dorado	0.0201	0.0201***	-0.1	99.8	496,262

Notes: Each row removes the indicated chain and re-estimates the preferred trend. The same-support baseline re-estimates the original measure on exactly the product-market-month cells that survive the omission. Panel A uses across-seller dispersion; Panel B uses chain-chain dispersion.

The Composition of Dispersion

A.8 Decomposing the Seller-Product Component

This subsection separates category-wide from within-category contributions to the seller-product trend. The baseline seller-product component accounts for approximately 90 percent of the fitted total-variance trend, but it can include a seller becoming relatively expensive for an entire category as well as finer differences among its products.

To distinguish these margins, let G_{igmt} denote the seller-product component defined in Section 4.1. For product i belonging to category c , define the seller-category component as

$$C_{gcmt} = \frac{1}{|\mathcal{I}_{gcmt}|} \sum_{j \in \mathcal{I}_{gcmt}} G_{jgmt},$$

where $\mathcal{I}_{gcmt}$ is the set of comparable products in category c carried by seller g in market m and month t . We then define the within-category seller-product component as

$$H_{igmt} = G_{igmt} - C_{gcmt}.$$

Hence $G_{igmt} = C_{gcmt} + H_{igmt}$. Applying the variance accounting identity from Section 4.1 partitions seller-product variance into category-wide variance, within-category variance, and twice their covariance.

Table A.12 reports component trends and divides each by the seller-product trend on that panel's support. Panel A uses the full sample: category, within-category, and covariance contributions are 46.3, 45.9, and 7.7 percent. Panel B first selects seller-category-market-month groups with at least two comparable products and constructs their category benchmarks. It then retains product-market-month cells with at least two surviving sellers. Benchmarks are not reconstructed after this final cell restriction: the two-product requirement refers to the benchmark-construction sample, and need not hold in every final group. Its shares use its own seller-product trend as denominator.

Table A.12: Category-Wide and Within-Category Contributions to Dispersion Growth

Component	Monthly trend	Standard error	Share of total trend (%)	Share of seller-product trend (%)
<i>Panel A: Full sample</i>				
Product–market–month observations: 648,318				
Total across-seller variance	0.4971***	0.0826	100.0	–
Seller-wide positions	0.0440***	0.0071	8.8	–
Seller-product positions	0.4480***	0.0780	90.1	100.0
Category-wide positions	0.2076***	0.0353	41.8	46.3
Positions within categories	0.2058***	0.0358	41.4	45.9
Twice category–within-category covariance	0.0346*	0.0204	7.0	7.7
Twice seller–seller-product covariance	0.0051	0.0093	1.0	–
<i>Panel B: At least two products per seller-category-market-month</i>				
Product–market–month observations: 576,377				
Seller-product positions	0.4528***	0.0835	–	100.0
Category-wide positions	0.1808***	0.0316	–	39.9
Positions within categories	0.2549***	0.0433	–	56.3
Twice category–within-category covariance	0.0170	0.0213	–	3.8

Notes: Panel A partitions the seller-product component on the full sample. Panel B constructs category benchmarks only in seller-category-market-month groups with at least two comparable products and then retains cells with at least two surviving sellers. Category positions include the focal product; Panel B is not a leave-one-product-out estimate. Seller-product shares use the seller-product trend in the corresponding panel as denominator; Panel B therefore reports only that partition.

A qualification is that the number of comparable products within a seller-category-market-month is sometimes small. The median is two products, and 27 percent of seller-category-market-month groups contain only one product, although these singleton groups account for only 13 percent of seller-product observations. When we restrict the analysis to groups containing at least two comparable products, 86 percent of the original seller-product observations remain. On this common multiproduct support, the seller-category component accounts for 39.9 percent of the seller-product trend, the within-category seller-product component accounts for 56.3 percent, and their covariance accounts for 3.8 percent.

Both margins contribute to the fitted widening; the within-category contribution is somewhat larger on multiproduct support. These shares locate variation in observed price positions; they do not identify common category shocks or synchronized adjustment.

A.9 Product-Composition Robustness of the Variance Decomposition

The baseline decomposition in Section 4 defines a seller’s overall price position as the equal-weighted average of its relative prices across the comparable products it carries in a market and month. A natural concern is that changes in the observed product mix could mechanically affect this seller-wide component.

We therefore construct an alternative seller-wide price position using a two-way fixed-effect model,

$$\tilde{p}_{igmt} = \alpha_{imt} + \delta_{gmt} + \varepsilon_{igmt},$$

where α_{imt} is a product–market–month fixed effect and δ_{gmt} is a seller–market–month price-position effect. The product–market–month fixed effects absorb the exact contemporaneous price level of every product, so the seller effects are estimated from overlapping product comparisons rather than from a simple average over the seller’s observed product mix.

Using these seller effects, we reconstruct the decomposition on the same seller-product observations used for the direct-average method. The seller-product component remains the residual between the same-good relative price and the adjusted seller-wide position, so the variance accounting identity in Section 4.1 holds exactly under both constructions.

Table A.13 compares component trends on identical observations and divides each by its total-variance trend. The seller-product share is 90.1 percent under direct averages and 90.0 percent under fixed effects; the seller-wide share changes from 8.8 to 10.9 percent and covariance offsets the difference. This is a robustness check for product-composition adjustment rather than an estimator of latent variance components.

The two-way fixed-effect exercise addresses differences in the contemporaneous product mix used to estimate seller-wide price positions. A separate concern is that the monitored

Table A.13: Product-Composition-Adjusted Dynamic Decomposition.

Component	Direct average		Two-way FE adjusted	
	Monthly trend	Std. Error	Monthly trend	Std. Error
Total variance	0.4971***	(0.0826)	0.4971***	(0.0826)
Seller component	0.0440***	(0.0071)	0.0543***	(0.0089)
Seller-product component	0.4480***	(0.0780)	0.4474***	(0.0776)
2 $\times$ covariance	0.0051	(0.0093)	-0.0046	(0.0100)
Seller share of total trend (%)	8.8		10.9	
Seller-product share of total trend (%)	90.1		90.0	
Observations	648,318		648,318	

Notes: Both columns use identical observations. The direct-average method uses the seller benchmark in Section 4.1; the two-way fixed-effect method estimates seller–market–month positions jointly with product–market–month effects. The seller-product component is the residual in both constructions, so the variance accounting identity in Section 4.1 holds exactly.

product basket itself changes over the long sample. To address this margin, we reconstruct the entire decomposition using only the 70 products observed during both the first and final 12 months. Importantly, the restriction is imposed before seller-wide price positions are constructed, so both the product–market cells entering the decomposition and the set of products used to characterize each seller are restricted to the long-lived product set.

Table A.14: Dynamic Morphology on a Fixed Long-Lived Product Set.

Sample	Total variance trend	Seller share (%)	Seller-product share (%)	Covariance share (%)	Observations
Full sample	0.4971***	8.8	90.1	1.0	648,318
Long-lived products	0.4505***	7.3	90.4	2.3	423,246

Notes: The restricted specification retains only products observed in both the first and final 12 months and reconstructs seller-wide positions using that long-lived set before computing the decomposition.

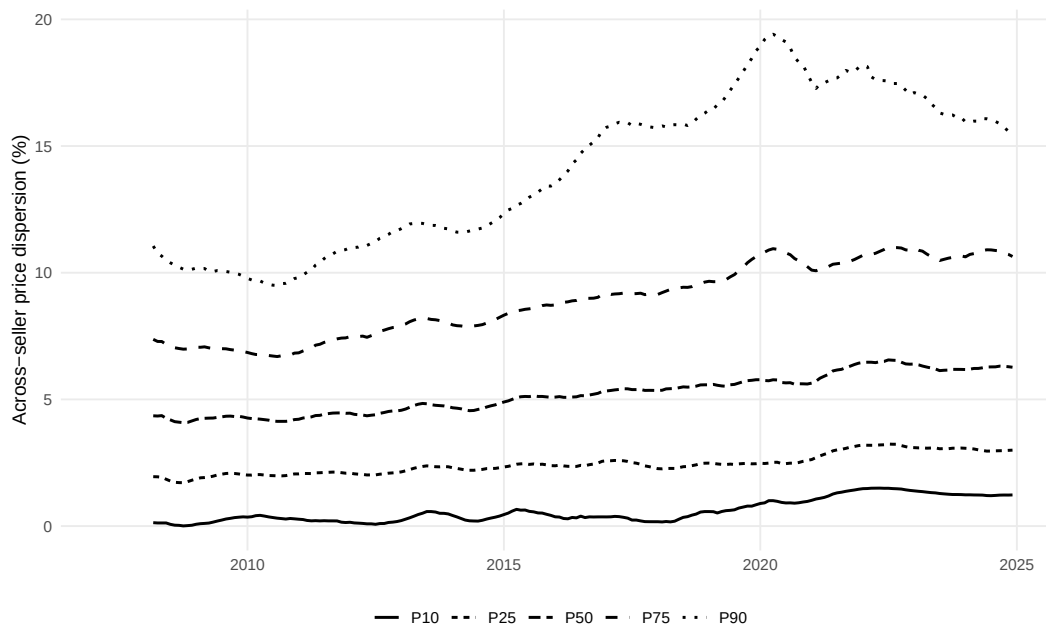
Table A.14 reports the rebuilt components using the restricted total trend as each share’s denominator. The total monthly variance trend falls from 0.4971 to 0.4505, while the seller-product share is 90.4 percent and the seller-wide share is 7.3 percent. Thus, the location of the fitted widening is similar when the monitored basket is restricted before constructing the benchmarks.

A.10 Sensitivity to Extreme Price Dispersion

Because the dynamic decomposition is based on variances, unusually large price gaps receive disproportionate weight. We therefore examine both whether the widening is visible throughout the distribution of across-seller dispersion and whether the component trends survive trimming extreme variance cells or winsorizing relative prices; only winsorization reconstructs the underlying components.

Figure A.5 reports 12-month moving averages of the seller-count-weighted 10th, 25th, 50th, 75th, and 90th percentiles of across-seller price dispersion. The widening is not confined to the upper tail. Between the first and final 12 months of the sample, median dispersion rises from 4.35 to 6.27 percent. The 25th percentile rises from 1.95 to 3.00 percent, while the 75th percentile rises from 7.37 to 10.64 percent. The increase is larger toward the upper tail—the 90th percentile rises by 4.45 percentage points—but the distribution shifts upward well below the most dispersed observations.

Figure A.5: Distribution of Across-Seller Price Dispersion over Time.



Notes: Twelve-month moving averages of seller-count-weighted monthly percentiles of across-seller dispersion, as defined in Section 3.3.

The upper tail nevertheless contributes substantially to the level of total variance. The

top 1 percent of product–market–month cells ranked by baseline total variance account for approximately 18 percent of weighted variance, and the top 5 percent account for approximately 44 percent. This makes it important to verify that the decomposition of the trend is not itself generated by these cells.

The tail restrictions either remove product–market–month cells above the pooled 99th or 95th percentile of baseline total variance or winsorize seller relative prices at the pooled 1st/99th or 5th/95th percentiles. The trimming specifications retain the original component values; after winsorization, relative prices are re-centered before the components are reconstructed so the variance decomposition remains exact.

Table A.15: Dynamic Morphology: Sensitivity to Extreme Price Dispersion.

Specification	Total variance trend	Seller share (%)	Seller-product share (%)	Covariance share (%)	Observations
Baseline	0.4971***	8.8	90.1	1.0	648,318
Drop top 1% variance cells	0.3605***	12.1	88.9	-1.0	641,851
Drop top 5% variance cells	0.1841***	22.9	89.4	-12.4	615,931
Winsorize seller relative prices 1/99	0.3462***	10.3	88.7	0.9	648,318
Winsorize seller relative prices 5/95	0.1499***	12.3	86.8	0.9	648,318

Notes: Top-cell restrictions drop observations above the pooled 99th or 95th percentile of baseline total variance. Winsorized specifications cap seller relative prices at the pooled 1st/99th or 5th/95th percentiles and then re-center them before reconstructing the decomposition. Shares use each row's total-variance trend as denominator.

Table A.15 reports the component trends under each restriction.

Restricting the tails reduces the magnitude of the estimated total-variance trend. Removing the top 1 percent of variance cells lowers the monthly trend from 0.4971 to 0.3605 squared percentage points, while removing the top 5 percent lowers it to 0.1841. The corresponding trends remain positive and highly statistically significant.

The seller-product shares are 88.9 and 89.4 percent after dropping the top 1 and 5 percent of variance cells, and 88.7 and 86.8 percent under the two winsorization rules. The component remains dominant, although tails materially affect total variance. These restrictions assess sensitivity to extremes; they do not identify whether an extreme obser-

vation is an economic price difference or a reporting error.

Price Histories and Adjustment

A.11 Observed-Spell Support and Selection

The reset-price decomposition requires observing the beginning of the current nominal-price spell. This restriction selects both product–market–month cells and sellers within those cells. Table A.16 separates these margins by rebuilding seller-wide and seller-product positions under three supports. Row A uses the full decomposition sample. Row B keeps only cells that survive in the identified-spell sample but restores all sellers observed in those cells. Row C then restricts those same cells to sellers whose current spell start is observed.

Table A.16: Why the Identified-Spell Support Has a Smaller Seller-Product Trend

Support	G trend	Std. Error	Full trend (%)	Change from previous	Obs.
A. Full cells + full sellers	0.4480***	0.0780	100.0	–	648,318
B. Spell-eligible cells + all sellers	0.3609***	0.0625	80.5	-19.5	542,963
C. Spell-eligible cells + identified sellers	0.2729***	0.0494	60.9	-19.6	542,963

Notes: Row A uses the full decomposition sample. Row B keeps the cells that survive in the identified-spell sample but restores all sellers in those cells. Row C then retains only sellers with an observed current-spell start. Thus A–B isolates cell selection and B–C within-cell seller selection. “Change from previous” is expressed as percentage points of the full-sample trend.

The full-sample seller-product trend is 0.4480 squared percentage points per month. Restricting only the set of cells lowers it to 0.3609, a reduction equal to 19.5 percent of the full trend. Restricting sellers within those same cells lowers the trend further to 0.2729, another 19.6 percent of the full trend. Thus, the smaller trend on the identified-spell support reflects both cell selection and seller selection within surviving cells. Among seller rows excluded from the identified-spell support, time gaps account for about two-thirds, initial observations for about one-fifth, and changes in the reporting branch set of chains for 13.3 percent.

A separate concern is simple left censoring at the beginning of the sample. Table A.17 therefore repeats the full-support versus identified-spell comparison after progressively removing the initial years. The gap narrows only modestly: the identified-spell trend remains roughly 36–39 percent below the corresponding full-support trend. By contrast, the reset share remains between 94 and 97 percent and the net-erosion share between 3 and 6 percent across these calendar windows. The support difference should therefore be treated as a limitation on extrapolating the magnitude of the reset decomposition to the full sample, not as evidence that the initial left-censoring period drives its internal allocation.

Table A.17: Reset-Price Decomposition: Support and Burn-In

Window	Full G	Spell G	Gap (%)	Reset share (%)	Net erosion share (%)	Coverage (%)
Full sample	0.4480	0.2729	-39.1	95.2	4.8	84.6
2008 onward	0.4586	0.2800	-39.0	94.6	5.4	85.0
2009 onward	0.4773	0.2965	-37.9	94.4	5.6	85.0
2010 onward	0.4906	0.3056	-37.7	94.8	5.2	85.0
2011 onward	0.4947	0.3114	-37.0	96.1	3.9	84.9
2012 onward	0.4981	0.3209	-35.6	96.6	3.4	85.1

Notes: Full G uses the canonical decomposition support; Spell G uses observations with an observed most-recent reset. Each post-2007 row re-estimates both trends on the same calendar window. Gap is their percentage difference; reset and net-erosion shares use Spell G as denominator. Coverage is the mean monthly share of seller-price rows with an observed spell start.

A.12 Robustness of the Reset-Price Decomposition

Table A.18 rebuilds the complete reset/erosion decomposition under alternative definitions of comparable price spells. Relaxing the requirement that a chain’s reporting branch set remain unchanged across adjacent months raises the observed-spell trend modestly but leaves 94.8 percent of it in reset positions. A separate filter removes exact one- and two-month nominal-price excursions that return to the pre-change nominal price before reconstructing spells. The filter replaces 4.2 percent of seller-price rows; overlapping ambiguous reversions are left unfiltered rather than resolved arbitrarily. The reset share

remains 94.9 percent.

The final rows restrict attention to retail chains. On this support, the reset-position trend is 0.2710 compared with an observed seller-product trend of 0.2742, so the last-reset reconstruction accounts for 98.8 percent of the fitted widening and the net-erosion trend is close to zero. Relaxing branch support within the chains-only sample yields a reset share of 98.6 percent. The result therefore does not depend on independent stores or on the exact treatment of changes in branch coverage.

Table A.18: Reset-Price Decomposition: Robustness

Specification	Observed G	Reset G	Net erosion	Reset share (%)	Net erosion share (%)	Obs.
Baseline: stable chain branch support	0.2729***	0.2598***	0.0131	95.2	4.8	542,963
Relax branch-support requirement	0.2867***	0.2719***	0.0148	94.8	5.2	555,565
Filter 1–2 month exact reversals	0.2680***	0.2544***	0.0136	94.9	5.1	533,703
Chains only	0.2742***	0.2710***	0.0032	98.8	1.2	430,742
Chains only: relaxed branch support	0.2848***	0.2808***	0.0041	98.6	1.4	440,295

Notes: The relaxed specification allows the reporting branch set of a chain to change across adjacent months. The short-reversal specification replaces exact one- and two-month nominal excursions that return to the pre-change price before reconstructing spells. Chains-only rows retain cells with at least two retail chains. Each row rebuilds the seller-wide and seller-product components before applying the reset/erosion decomposition.

A.13 Nominal Adjustment Timing and Product Composition

Table A.19 separates the extensive and intensive margins of nominal monthly adjustment. The frequency and conditional-size statistics use January 2008–December 2024; observed-price-age statistics begin in January 2009. Reported means are averages of monthly statistics. The first three-year windows are therefore 2008–2010 for frequency and size and 2009–2011 for age; the final window is 2022–2024. Over 2008–2024, the probability of any reset has no statistically significant linear trend. The probability of an increase declines weakly, while the probability of a decrease is essentially trendless. Conditional adjustment sizes move differently: the mean increase rises from about 7.1 to 9.1 percentage

points between the first and last three-year windows, and the mean absolute decrease rises from about 8.2 to 11.1.

The upper part of the observed price-age distribution shifts outward. These are ages of prices currently in force, not completed spell durations. Longer spells are more likely to be represented in this stock, and initial spells with unknown beginnings are excluded. The median changes little, but the 75th and 90th percentiles increase, as do the shares of observed prices aged at least 6 and 12 months. These age measures use only spells whose beginning is observed and begin in 2009 to reduce initial censoring. The combination of a reset probability with no significant 2008–2024 trend and a growing upper tail of price age implies that adjustment timing cannot be summarized by a uniform shift in stickiness.

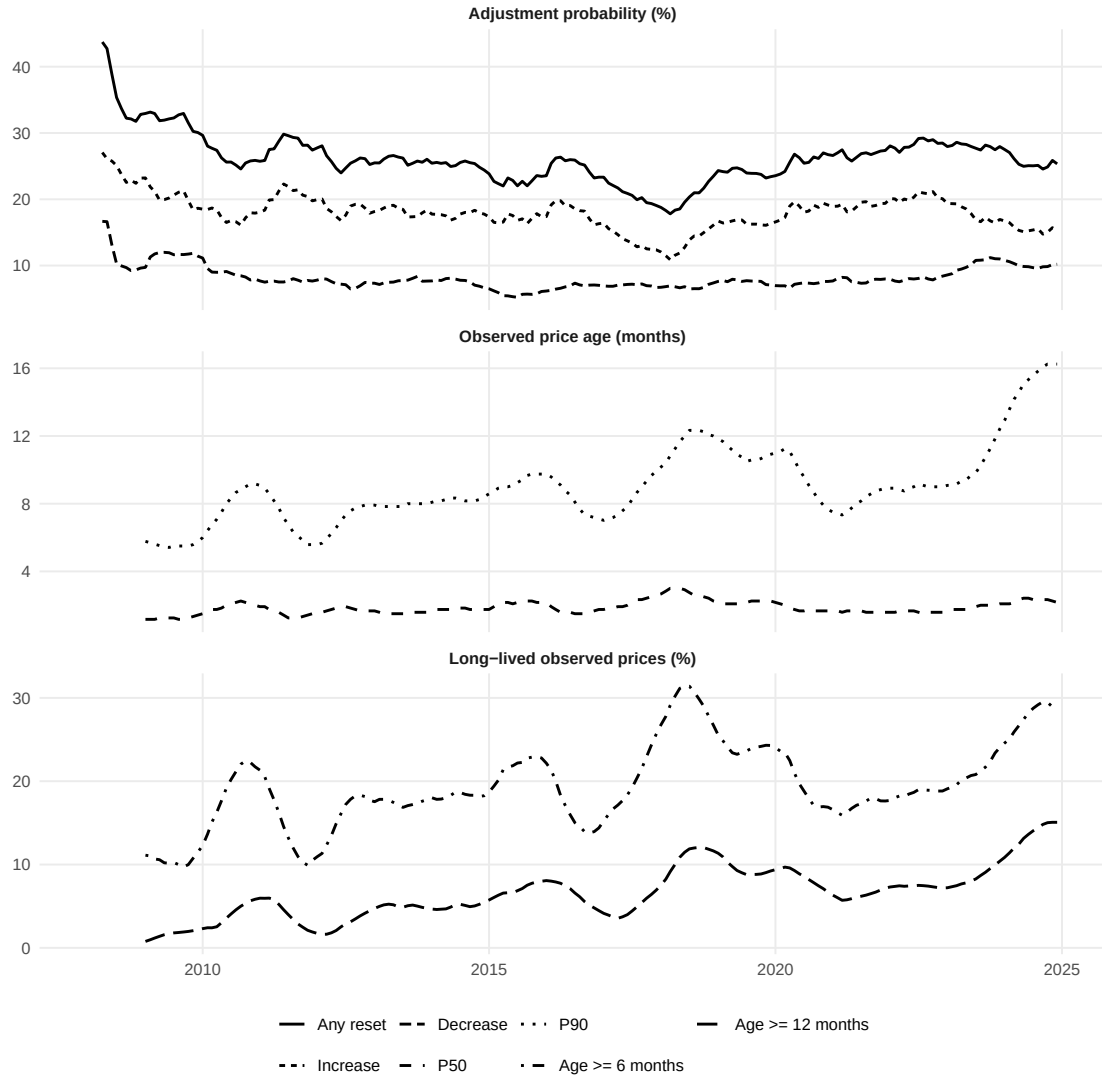
Table A.19: Margins of Nominal Price Adjustment

Measure	Mean	First 3y	Last 3y	Monthly trend	NW(12) SE	<i>p</i> -value
Monthly reset probability	25.91	29.58	27.26	-0.0137	0.0127	0.279
Probability of increase	17.81	19.92	17.40	-0.0152*	0.0085	0.073
Probability of decrease	8.10	9.67	9.87	0.0015	0.0081	0.854
Mean increase size	8.34	7.05	9.14	0.0114***	0.0018	< 0.001
Mean absolute decrease size	10.53	8.21	11.09	0.0173***	0.0040	< 0.001
Median observed price age	1.88	1.64	1.94	0.0020	0.0013	0.115
P75 observed price age	4.58	3.75	5.28	0.0089**	0.0035	0.012
P90 observed price age	9.30	6.83	12.56	0.0326***	0.0098	0.001
Share with age ≥ 6 months	19.83	14.57	23.81	0.0520***	0.0180	0.004
Share with age ≥ 12 months	6.96	3.30	10.88	0.0467***	0.0099	< 0.001

Notes: Adjustment probabilities use consecutive comparable economic-seller monthly modal prices; chain transitions require unchanged branch support. Increase and decrease probabilities are unconditional shares of comparable transitions. Conditional sizes are 100 times the log increase and 100 times the absolute log decrease. Price-age statistics use spells with observed beginnings and start in 2009. Trends use monthly aggregates, calendar-month indicators, and Newey–West standard errors with 12 lags.

Product-composition robustness. The monitored product list underwent a major revision in 2010–2011, and products can differ substantially in their adjustment frequencies and durations. Table A.20 therefore repeats the stickiness trends for the 70 products observed in both the first and final 12 months and for samples beginning in 2012. Spells are constructed using the full available history before imposing the calendar restrictions,

Figure A.6: Nominal Price Adjustment and the Distribution of Price Ages.



Notes: Twelve-month moving averages of the adjustment and observed-price-age measures defined in Table A.19. Price age uses spells with observed beginnings; chain transitions require unchanged branch support.

so a price already in place in 2012 retains its observed age.

The longer upper tail survives all of these restrictions. Among long-lived products, the trend in the 90th percentile of price age remains positive and significant, and the share of observed prices aged at least 12 months rises significantly. Starting the sample in 2012 gives the same qualitative result. In the 2012-onward long-lived sample, adjustment frequency itself rises significantly while the upper tail of observed price age continues to lengthen. Thus, the longer upper tail is not explained by the 2010–2011 basket revision

Table A.20: Nominal Price Stickiness: Product-Composition Robustness

Measure	Full sample	Long-lived products	2012 onward all products	2012 onward long-lived
Monthly reset probability	-0.0137 (0.0127)	-0.0093 (0.0143)	0.0226* (0.0119)	0.0366*** (0.0125)
P50 observed price age	0.0020 (0.0013)	0.0015 (0.0014)	0.0012 (0.0016)	-0.0007 (0.0016)
P75 observed price age	0.0089** (0.0035)	0.0060* (0.0033)	0.0076* (0.0044)	0.0038 (0.0042)
P90 observed price age	0.0326*** (0.0098)	0.0261*** (0.0095)	0.0339*** (0.0129)	0.0265** (0.0131)
Share with age ≥ 6 months	0.0520*** (0.0180)	0.0398** (0.0182)	0.0398** (0.0194)	0.0237 (0.0202)
Share with age ≥ 12 months	0.0467*** (0.0099)	0.0388*** (0.0097)	0.0472*** (0.0130)	0.0386*** (0.0133)

Notes: Each cell reports the monthly trend with its Newey–West(12) standard error. Long-lived products are observed in both endpoint windows; 2012-onward samples begin after the basket revision. Spells are constructed from the full available history before calendar restrictions, so spells already in progress in 2012 retain their observed age. Other construction details follow Table A.19.

alone. The post-2012 aggregate increase in adjustment frequency coexists with a growing share of old prices; these moments do not identify changes in individual adjustment rules.

A.14 Within-Seller Price-Change Moments

To assess whether both directions emphasized by the model have empirical counterparts in observed adjustment, we apply the same common–relative transformation to pairs of monthly nominal price changes. For two distinct products i and j observed for the same economic seller, neighborhood, and month, let x_i and x_j denote their log price changes, multiplied by 100, and define

$$a = \frac{x_i + x_j}{2}, \quad d = \frac{x_i - x_j}{2}.$$

The identity $a^2 + d^2 = (x_i^2 + x_j^2)/2$ separates squared pairwise adjustment into common and relative directions. Pair moments give each seller–neighborhood–month equal total weight. For retail chains, adjacent months are retained only when the reporting branch set is unchanged, as in the baseline price-spell construction.

Table A.21: Within-Seller Price-Change Moments

Moment	All economic sellers	Retail chains
Common-direction share of squared pairwise changes (%)	51.84	51.90
Relative-direction share of squared pairwise changes (%)	48.16	48.10
Marginal product adjustment probability (%)	26.48	27.50
Both products adjust (% of product pairs)	8.67	9.27
Joint adjustment under independence (%)	7.01	7.56
Other product adjusts conditional on one adjusting (%)	32.75	33.71
Opposite signs conditional on both adjusting (%)	38.40	38.54

Notes: Pair construction and the a, d transformation are defined in Section A.14. The marginal adjustment probability is computed on the pair sample as $P(A_i) = P(A_i = A_j = 1) + \frac{1}{2}P(A_i \neq A_j)$; it differs from the 25.91 percent overall reset probability in Table A.19 because support and weighting differ. The independence benchmark is $P(A_i)^2$.

Observed joint adjustment exceeds the independence benchmark only modestly, indicating partial rather than complete synchronization. These descriptive moments are not used to calibrate the model or recover desired-price gaps.

Online Appendix B. Disaggregated Montevideo CPI: Sources, Linking, and Coverage

This exercise compares inflation in official Montevideo consumer-price categories with dispersion across sellers of the same DGC product. The two series measure different objects: the CPI tracks changes in a category price index over time, whereas DGC dispersion measures price differences across sellers within a product, market, and month. We link them at the category–month level to study their comovement. Neither the CPI category index nor its underlying basket is identical to the set of branded products monitored by DGC.

Sources and reference periods. We use two disaggregated Montevideo CPI releases from Uruguay’s Instituto Nacional de Estadística (INE). The legacy series has December 2010 = 100, with a consumption basket derived from the 2005–2006 Encuesta Nacional de Gastos e Ingresos de los Hogares (ENGIH) (Instituto Nacional de Estadística, 2011). The new series has October 2022 = 100, with a basket derived from the 2016–2017 ENGIH and expenditure weights updated by prices to October 2022 (Instituto Nacional de Estadística, 2022). The classification also changes from an adaptation of COICOP 1999 to COICOP 2018. Rebasing therefore involves changes in weights, coverage, and classification, not merely a new index reference period. Both releases provide regional indices; we use Montevideo throughout. The legacy release provides division, group, and class indices, while the new release includes finer subclass and product indices and official weights.

The construction reads index levels from the two releases, rather than their published percentage-change or incidence columns. Official codes are retained as character strings, including leading zeros, and dates are standardized to calendar months. The legacy series starts with its December-2010 normalization of 100. October 2022 is the common linking month. Conflicting duplicate code–month observations and missing months within a linked series trigger an error rather than being averaged or interpolated.

Goods covered and category correspondence. Table B.1 reports the fixed correspondence for 42 DGC categories, covering food, non-alcoholic and alcoholic beverages, household cleaning goods, and personal-care and hygiene products. These are categories, not 42 individual branded products, and they do not constitute the full consumer basket: services, housing, and other major CPI components are outside this correspondence. The DGC panel is not expenditure weighted and does not provide the transaction quantities needed to reproduce a scanner-data inflation index.

The 42 categories map to 13 distinct legacy CPI series and 39 distinct new CPI series. Several categories consequently share an inflation series, especially before the rebasing. For example, the legacy bread-and-cereals class is assigned to several DGC cereal products, while the new classification distinguishes rice, flour, pasta, and other narrower concepts. Under the new classification, 27 category matches are labelled exact, 13 close/broader, and two broad. “Exact” refers to the correspondence between category concepts, not to identical brands, package sizes, establishments, or transaction weights in the two sources. The codes and match classifications are fixed before estimation. To avoid repeating the coarser legacy labels, Table B.1 groups categories by IPC 2010 series and reports the IPC 2022 code, concept, and match for each DGC category.

Linking the two index vintages. Let I_{ct}^{10} and I_{ct}^{22} denote the old and new official indices assigned to DGC category c , and let T denote October 2022. We construct

$$I_{ct} = \begin{cases} I_{ct}^{10}, & t \leq T, \\ I_{ct}^{22} \frac{I_{cT}^{10}}{I_{cT}^{22}}, & t > T. \end{cases}$$

Thus, each new index is rescaled by a category-specific linking factor to match the old index in October 2022. The old release supplies observations through that month and the new release supplies observations from November 2022 onward. The November growth rate is exactly $\log I_{c,\text{Nov}}^{22} - \log I_{c,\text{Oct}}^{22}$: the linking factor cancels. A change from a base of 100 in December 2010 to a base of 100 in October 2022 therefore does not itself generate

measured inflation.

This procedure preserves the growth rates of each source on its respective segment; it does not reconstruct a constant historical basket. In particular, linking a broad legacy class to a narrower new concept removes a level discontinuity but does not remove the change in goods coverage or official weights. The greater disaggregation after October 2022 must therefore be kept in mind when interpreting changes in the inflation–dispersion association across the two segments. Twelve-month changes spanning the link also combine growth rates from both classifications.

Inflation measures and units. From the linked index we calculate monthly log inflation, $100(\log I_{ct} - \log I_{c,t-1})$, annualized monthly log inflation,

$$\pi_{ct} = 1200(\log I_{ct} - \log I_{c,t-1}),$$

and twelve-month log inflation, $100(\log I_{ct} - \log I_{c,t-12})$. Annualized monthly inflation, used in the baseline following Sheremirov (2020), is twelve times the one-month log change; it is not the observed change over the previous twelve months. With the December-2010 index normalization, monthly inflation can first be calculated for January 2011 and twelve-month inflation for December 2011. Consequently, this CPI exercise does not cover the earliest years of the full DGC price panel.

Constructing dispersion and merging the sources. For each DGC product–market–month, dispersion is 100 times the sample standard deviation of log monthly modal posted prices across physical stores, requiring at least two stores with valid positive prices. Dispersion remains at this level: we do not average it across products within categories before estimation. One unit therefore corresponds to one percentage point of the standard deviation of log prices. Monthly modal prices should not be interpreted as regular prices purged of all promotions.

Each observation is matched to official Montevideo inflation using its DGC category

and calendar month. All neighborhoods receive the same inflation rate for a given category and month, and categories mapped to the same official CPI series share that rate. Repeating inflation across products and neighborhoods does not create additional independent CPI series. Official expenditure weights enter INE’s published indices but are not used to weight DGC products or to construct a new aggregate CPI.

The estimation period is January 2011–December 2024. Observations require valid dispersion and matched finite inflation; specifications with lagged controls also require the corresponding monthly history. The 42-category crosswalk describes potential matching coverage, rather than the number of individual products retained in every regression. Appendix Section [A.4](#) presents the regression specifications, estimation samples, and interpretation. This appendix documents the construction of the inflation series used there.

Table B.1: Matching DGC Categories to the Official Montevideo CPI

DGC category	IPC 2022 code	IPC 2022 concept	Match
<i>IPC 2010 0111: Pan y cereales</i>			
Arroz	01111201	Arroz	Exact
Fideos al huevo	01115001	Fideos secos	Close/Broader
Fideos semolados	01115001	Fideos secos	Close/Broader
Galletas al agua	01113109	Galletas saladas	Close/Broader
Harina de maíz	01112601	Harina de maíz	Exact
Harina de trigo 000	01112101	Harina de trigo	Close/Broader
Harina de trigo 0000	01112101	Harina de trigo	Close/Broader
Pan de molde	01113105	Pan blanco envasado en rebanadas	Exact
<i>IPC 2010 0112: Carne</i>			
Hamburguesas	01125902	Hamburguesa de carne preparada para cocinar	Exact
<i>IPC 2010 0114: Leche, queso y huevos</i>			
Huevos	01148101	Huevos de gallina	Exact
Queso rallado	01145002	Queso rallado	Exact
Yogur	01146	Yogur y productos similares	Close/Broader
<i>IPC 2010 0115: Aceites y grasas</i>			
Aceite de girasol	01151101	Aceite de girasol	Exact
Aceite de maíz	01151	Aceites vegetales	Broad
Aceite de soja	01151401	Aceite de soja	Exact
Manteca	01152101	Manteca	Exact
Margarina	0115	Aceites y grasas	Broad
<i>IPC 2010 0117: Legumbres y hortalizas</i>			
Arvejas	01179201	Arvejas enlatadas	Exact
<i>IPC 2010 0118: Azúcar, mermelada, miel, chocolate y dulces de azúcar</i>			
Azucar	01181101	Azúcar blanco refinada	Exact
Dulce de Leche	01183201	Dulce de leche	Exact
Helado	01186001	Helados	Exact

Continued on next page

Table B.1 – continued

DGC category	IPC 2022 code	IPC 2022 concept	Match
Mermelada	01183901	Mermeladas y jaleas	Exact
<i>IPC 2010 0119: Productos alimenticios n.e.p.</i>			
Mayonesa	01193903	Mayonesa, salsa golf, mostaza, ketchup, salsa de soja	Close/Broader
Pulpa de tomate	01193902	Pulpa de tomate	Exact
Sal	01193101	Sal	Exact
<i>IPC 2010 0121: Café, té y cacao</i>			
Café	01220	Café y sucedáneos del café	Close/Broader
Cocoa	01185301	Cocoa en polvo	Exact
Te	01230201	Té en saquitos	Exact
Yerba	01230501	Yerba mate	Exact
<i>IPC 2010 0122: Aguas minerales, refrescos, jugos de frutas y de legumbres</i>			
Agua	01250001	Agua	Exact
Refresco	01260	Refrescos	Close/Broader
<i>IPC 2010 0212: Vino</i>			
Vino	02121001	Vino	Exact
<i>IPC 2010 0213: Cerveza</i>			
Cerveza	02130001	Cerveza	Exact
<i>IPC 2010 0561: Bienes para el hogar no duraderos</i>			
Detergente	05611003	Detergente	Exact
Hipoclorito	05611002	Blanqueador líquido	Close/Broader
Jabon de ropa	05611001	Jabones de limpieza en polvo y en barra, suavizante	Close/Broader
Jabon en polvo	05611001	Jabones de limpieza en polvo y en barra, suavizante	Close/Broader
<i>IPC 2010 1213: Otros aparatos, artículos y productos para la atención personal</i>			
Champú	13120005	Shampoo, acondicionador, savia, cremas para el cabello	Close/Broader
Desodorante	13120006	Desodorantes y antitranspirantes	Exact
Jabon de tocador	13120002	Jabón de tocador, líquido o en barra, gel de ducha	Exact
Papel higienico	13120007	Papel higiénico	Exact
Pasta de dientes	13120004	Pasta dental	Exact

Online Appendix C. Derivations and Numerical Verification

C.1 Demand, Resource Feasibility, and Reset Prices

The adjustment-cost technology is the finite two-product analogue of Bonomo, Carvalho, Kryvtsov, Ribon, and Rigato (2023). Their partial-synchronization model combines a common fixed repricing cost with an additional cost proportional to the fraction of products adjusted. With two products, we write this as $K\mathbf{1}\{J_g \neq \emptyset\} + c|J_g|/2$, so the individual term is the two-product counterpart of their proportional repricing cost.

The household solves $\max\{\log C - \chi N\}$ subject to $PC = WN + \Pi$, taking wages, profits, and prices as given. Expenditure minimization for $C = 2\sqrt{C_1 C_2}$ allocates half of spending to each product and gives $P = \sqrt{P_1 P_2}$. CES minimization within products gives

$$C_i = \left(\int c_{gi}^{(\varepsilon-1)/\varepsilon} dg \right)^{\varepsilon/(\varepsilon-1)}, \quad c_{gi} = \frac{M}{2} P_i^{\varepsilon-1} p_{gi}^{-\varepsilon}.$$

The labor first-order condition is $W/P = \chi C$. Imposing the aggregate expenditure anchor gives $W = \chi M$. For adjustment subset J_g , define

$$\kappa(J_g) = K\mathbf{1}\{J_g \neq \emptyset\} + \frac{c}{2}|J_g|.$$

Total labor and profits are

$$N = \int_0^1 \left[\sum_i \frac{c_{gi}}{A_{gi}} + \kappa(J_g) \right] dg, \quad \Pi = \int_0^1 \left[\sum_i (p_{gi} - W/A_{gi})c_{gi} - W\kappa(J_g) \right] dg.$$

Hence $WN + \Pi = \int \sum_i p_{gi} c_{gi} dg = M$. Repricing costs use real resources. Maximizing an adjusted product's operating-profit contribution gives the unique positive reset price

$$p_{gi}^* = \frac{\varepsilon W}{(\varepsilon - 1)A_{gi}}.$$

Because operating profit is separable across products conditional on aggregate indices, the firm chooses which products to reset through the discrete subset problem in equation (17). The two types used numerically are masses of atomistic firms, not two strategic sellers.

C.2 Exact Product-Level Gains and the Local Selection Rule

Write $x_i = \log(p_{gi}^0/p_{gi}^*)$. Holding aggregate indices fixed, the gross gain in labor units from resetting product i is

$$L_{gi}(x_i) = \frac{M}{2W} P_i^{\varepsilon-1} (p_{gi}^*)^{1-\varepsilon} b(x_i), \quad b(x) = \frac{1}{\varepsilon} - e^{(1-\varepsilon)x} + \frac{\varepsilon-1}{\varepsilon} e^{-\varepsilon x}. \quad (27)$$

Here $b(0) = b'(0) = 0$ and $b''(0) = \varepsilon - 1$. At the normalized symmetric reference point $P_i = p_{gi}^* = M = 1$ and $W = (\varepsilon - 1)/\varepsilon$,

$$L_{gi}(x_i) = \frac{\varepsilon}{4} x_i^2 + O(|x_i|^3).$$

With two products, the gross gain from a subset J is the sum of the corresponding product gains, whereas the labor cost is $K + c|J|/2$ when J is nonempty. Ignoring third-order terms, define

$$q_i = \frac{\varepsilon}{4} x_i^2, \quad s_i = q_i - \frac{c}{2}.$$

Conditional on paying the common cost, product i is included if $s_i > 0$. The firm changes at least one price if and only if

$$\sum_i (s_i)_+ > K,$$

which yields equation (20). In (a, d) coordinates, $x_1 = a + d$ and $x_2 = a - d$, so the local inaction set is

$$\left[\frac{\varepsilon}{4} (a + d)^2 - \frac{c}{2} \right]_+ + \left[\frac{\varepsilon}{4} (a - d)^2 - \frac{c}{2} \right]_+ \leq K.$$

The individual cost therefore replaces the circular inaction set of a pure joint-cost model with the kinked boundary shown in Figure 6.

C.3 Proof of Proposition 1

Initially, every product gap has magnitude r . The condition $\varepsilon r^2/4 < \tau = c/2$ implies that no product is individually worth repricing, so both economies are initially inactive.

After the nominal shock, the cheap C type has product gaps $(-r - \delta, -r - \delta)$ and therefore two gross gains equal to H . Since $H > \tau$ and $2(H - \tau) > K$, it pays the common cost and reprices both products. The expensive C type has two gross gains equal to $L < \tau$ and remains inactive.

Each R type has one gap with gross gain H and one with gross gain L . The low-gain product is not worth repricing because $L < \tau$. The remaining product generates net product-level surplus $H - \tau < K$, which is insufficient to cover the common cost. Both R types therefore remain inactive. The resulting price and consumption responses are the same first-order expressions reported in Proposition 1. The neighboring local regimes follow from the same rule. If $K < H - \tau$, the cheap C type resets both high-gap products and each R type resets its single high-gap product; if $K > 2(H - \tau)$, even the cheap C type's two net product surpluses are insufficient to cover the common cost, so no type reprices. These are local quadratic statements; exact CES cutoffs can differ because the product price indices are endogenous. The result is a finite threshold-crossing comparison under the local approximation; it is not an infinitesimal elasticity or a stationary dynamic result.

C.4 Exact Equilibrium Checks

The numerical example uses

$$\varepsilon = 6, \quad \chi = \frac{5}{6}, \quad K = .0045, \quad c = .003, \quad r = .02, \quad \delta = .035.$$

Repricing one product costs $K + c/2 = .0060$ units of labor and repricing both costs $K + c = .0075$.

The verification code evaluates every subset $J \in \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$ for each type,

computes the implied CES indices, and checks all discrete profit comparisons. It also uses global bounds based on the fact that each CES product index lies between the smallest and largest feasible price for that product.

In both initial economies, every product-specific gross reset gain is bounded above by $0.000714 < c/2 = .0015$, so no product is worth repricing. After the nominal shock in C, each product of the cheap type has gross gain bounded below by $0.004228 > c/2$ and

$$2(0.004228 - 0.0015) = 0.005455 > K = .0045,$$

so the type strictly prefers to reset both prices. For the expensive C type, each product-specific gain is bounded above by $0.000357 < c/2$, so it remains inactive.

After the shock in R, each type has one high-gap and one low-gap product. The high-gap product's gross gain is bounded above by 0.005566 , while the low-gap product's gain is bounded above by $0.000357 < c/2$. Thus even the best possible one-product surplus satisfies

$$0.005566 - 0.0015 = 0.004066 < K = .0045,$$

so neither R type pays the common cost. Combining this lower bound with the cheap-C lower bound above gives the sufficient strict window

$$0.004066 < K < 0.005455,$$

which contains the reported value $K = .0045$. These inequalities are strict for every feasible product index and therefore also rule out partial adjustment within either type in the reported C-versus-R experiment. The local regime boundaries in Section 6 are first-order summaries; the exact CES boundaries need not coincide with them because product price indices are endogenous.

The code checks expenditure, the household budget, resource feasibility, labor supply, CES aggregation, and reset-price first-order conditions. The intended action profile

is the unique pure type-level equilibrium in every reported state. Initial product-price distributions coincide exactly across C and R. The parameter choices are illustrative, not calibrated to the empirical synchronization moments.

C.5 Relative Shocks and Conditional Full Pass-Through

Set $A_{gi} = \exp(-z_{gi})$, with $\mathbf{z}_1 = (-s, s)$ and $\mathbf{z}_2 = (s, -s)$, inherited prices equal to one, and $s = .08$. Under the normalization, $p_{gi}^* = M \exp(z_{gi})$.

At $M = 1$, the smallest global lower bound on a product-specific gross reset gain is $0.005815 > c/2 = .0015$. The sum of the two lower bounds net of individual costs exceeds the common cost:

$$0.007200 + 0.005815 - c = 0.010015 > K = .0045.$$

Thus every firm strictly prefers to reprice both products. Holding productivities and inherited prices fixed and setting $M = e^{.035}$, the smallest product-specific lower bound is still $0.002580 > c/2$, and

$$0.002580 + 0.013740 - c = 0.013320 > K.$$

All firms again strictly prefer to reprice both products. These inequalities apply for every feasible product index.

Exact scaling argument. More generally, fix productivities and consider $M_1 = e^\delta M_0$. If all firms reset both prices in both equilibria, then $W_1 = e^\delta W_0$ and

$$p_{gi,1} = e^\delta p_{gi,0}, \quad P_{i,1} = e^\delta P_{i,0}, \quad P_1 = e^\delta P_0.$$

The first equality follows from the reset-price rule, and the index equalities follow from homogeneity. Here the subscripts 0 and 1 denote counterfactual equilibria, not successive

periods. Product demand satisfies

$$\frac{c_{gi,1}}{c_{gi,0}} = \frac{M_1}{M_0} \left(\frac{P_{i,1}}{P_{i,0}} \right)^{\varepsilon-1} \left(\frac{p_{gi,1}}{p_{gi,0}} \right)^{-\varepsilon} = e^{\delta+\delta(\varepsilon-1)-\delta\varepsilon} = 1.$$

Consequently, $C_1/C_0 = (M_1/M_0)/(P_1/P_0) = 1$. Productivities are fixed, so production labor is unchanged. Repricing labor equals $K + c = .0075$ in both equilibria because every firm changes both prices; total employment is therefore unchanged as well. In the numerical example, the incremental nominal shock raises the price index by exactly 3.5 log points and leaves consumption and employment unchanged.

Universal adjustment is a condition verified for the stated experiment, not an assumption valid for every nominal shock. The technology itself permits partial adjustment in other states: for example, under the local approximation a sufficiently large gap in one product combined with a small gap in the other can justify paying $K + c/2$ for a one-product reset while leaving the second price unchanged. The role of this experiment is only to establish that relative repricing can coexist with full nominal pass-through when both product-level gaps activate adjustment.

C.6 Relative Repricing and the Aggregation of Prices

A separate measurement result follows from the relative-cost experiment at $M = 1$. Once all firms reset, each product across sellers has prices e^{-s} and e^s with equal mass. Its mean log price is zero and its log-price variance is s^2 . The arithmetic mean and CES index are

$$\bar{p}_i = \cosh(s), \quad P_i = P = [\cosh((\varepsilon - 1)s)]^{-1/(\varepsilon-1)}.$$

Around symmetry,

$$\log \bar{p}_i = \frac{s^2}{2} + O(s^4), \quad \log P = -\frac{\varepsilon - 1}{2} s^2 + O(s^4).$$

For $s = .08$, the arithmetic mean rises by 0.320% and the index falls by 1.559 log points relative to the initial symmetric reference with all prices and productivities equal to one. Mean log prices remain exactly zero. Thus relative repricing can widen each product's distribution without moving its logarithmic center, but this does not establish neutrality of the economic price index.

This comparison changes real productivities; it is distinct from the nominal counterfactual above, which holds them fixed. Desired heterogeneity is generated by relative costs in this example, not by an identified competitive-repositioning mechanism. No external CPI deflator is introduced here.

Online Appendix D. Product List and Product Characteristics

Table D.1: Products, Presentations, and Sample Entry Dates

Product	Brand	Specification	CPI share (%)	Owner / merger	Sample start (merger)
Beer	Patricia	0.96 L	0.38	FNC	2007/04
Beer	Pilsen	0.96 L	0.38	FNC	2007/04
Beer	Zillertal	1 L	0.38	FNC	2010/11
Wine	Faisán	1 L	0.80	Grupo Traversa	2007/04
Wine	Santa Teresa Clasico	1 L	0.80	Santa Teresa SA	2007/04
Wine	Tango	1 L	0.80	Almena	2007/04
Carbonated Soft Drink	Coca Cola	1.5 L	1.12	Coca Cola	2007/04
Carbonated Soft Drink	Nix	1.5 L	1.12	Milotur (CCU)	2007/04
Carbonated Soft Drink	Pepsi	1.5 L	1.12	Pepsi	2010/11
Still water	Matutina	2 L	0.81	Salus	2007/04
Still water	Nativa	2 L	0.81	Milotur (CCU)	2007/04
Still water	Salus	2.25 L	0.81	Salus	2007/04
Bread Loaf	Los Sorchantes	0.33 Kg	0.06	Bimbo / Los Sorchantes	2010/11 (2011/04)
Bread Loaf	Bimbo	0.33 Kg	0.06	Bimbo	2010/11
Bread Loaf	Pan Catalán	0.33 Kg	0.06	Bimbo	2010/11
Brown eggs	Super Huevo	1/2 dozen	0.46	Super Huevo	2010/11
Brown eggs	El Jefe	1/2 dozen	0.46	El Jefe	2010/12
Brown eggs	Prodhin	1/2 dozen	0.46	Prodhin	2007/07
Butter	Calcar	0.2 Kg	0.23	Calcar	2007/04
Butter	Conaprole sin sal	0.2 Kg	0.23	Conaprole	2007/04
Butter	Kasdorf	0.2 Kg	0.23	Conaprole	2010/11
Cacao	Copacabana	0.5 Kg	0.08	Nestlé	2007/04
Cacao	Vascolet	0.5 Kg	0.08	Nestlé	2007/06
Coffee	Aguila	0.25 Kg	0.14	Nestlé	2007/04
Coffee	Chana	0.25 Kg	0.14	Nestlé	2007/04
Coffee	Saint	0.25 Kg	0.14	Saint Hnos	2010/11
Corn Oil	Delicia	1 L	n/i	Cousa	2010/11
Corn Oil	Río de la Plata	1 L	n/i	Soldo	2010/11

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Table D.1 continued

Product	Brand	Specification	CPI share (%)	Owner / merger	Sample start (merger)
Corn Oil	Salad	1 L	n/i	Nidera	2010/11
<i>Dulce de leche</i>	Conaprole	1 Kg	0.14	Conaprole	2007/04
<i>Dulce de leche</i>	Los Nietitos	1 Kg	0.14	Los Nietitos	2007/04
<i>Dulce de leche</i>	Manjar	1 Kg	0.14	Manjar	2007/04
Flour (corn)	Gourmet	0.4 Kg	n/i	Deambrosi	2010/11
Flour (corn)	Presto Pronta Arcor	0.5 Kg	n/i	Arcor	2010/11
Flour (corn)	Puritas	0.45 Kg	n/i	Molino Puritas	2010/11
Flour 000 (wheat)	Cañuelas	1 Kg	0.21	Molino Cañuelas	2010/11
Flour 000 (wheat)	Cololó	1 Kg	0.21	Distribuidora San José	2010/11
Flour 0000 (wheat)	Cañuelas	1 Kg	0.21	Molino Cañuelas	2007/04
Flour 0000 (wheat)	Cololó	1 Kg	0.21	Distribuidora San José	2007/04
Flour 0000 (wheat)	Primor	1 Kg	0.21	Molino San José	2010/11
Grated cheese	Conaprole	0.08 Kg	0.16	Conaprole	2007/04
Grated cheese	Artesano	0.08 Kg	0.16	Artesano	2010/11
Grated cheese	Milky	0.08 Kg	0.16	Milky	2007/04
Deodorant	Axe Musk	0.105 Kg	0.34	Unilever	2010/11
Deodorant	Dove Original	0.113 Kg	0.34	Unilever	2010/11
Deodorant	Rexona Active Emotion	0.100 Kg	0.34	Unilever	2010/11
Hamburger	Burgy	0.2 Kg	n/i	Schneck	2010/11
Hamburger	Paty	0.2 Kg	n/i	Sadia Uruguay	2010/11
Hamburger	Schneck	0.2 Kg	n/i	Schneck	2010/11
Ice Cream	Conaprole	1 Kg	0.22	Conaprole	2010/11
Ice Cream	Crufi	1 Kg	0.22	Crufi	2010/11
Ice Cream	Gebetto	1 Kg	0.22	Conaprole	2010/11
Margarine	Flor	0.2 Kg	n/i	Cousa	2010/11
Margarine	Doriana nueva	0.25 Kg	n/i	Unilever	2007/04
Margarine	Primor	0.25 Kg	n/i	Cousa	2007/04
Mayonnaise	Fanacoa	0.5 Kg	0.21	Unilever	2007/04
Mayonnaise	Hellmans	0.5 Kg	0.21	Unilever	2007/04
Mayonnaise	Uruguay	0.5 Kg	0.21	Unilever	2007/04
Noodles	Cololo	0.5 Kg	0.43	Distribuidora San José	2007/07
Noodles	Adria	0.5 Kg	0.43	La Nueva Cerro	2007/07

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Table D.1 continued

Product	Brand	Specification	CPI share (%)	Owner / merger	Sample start (merger)
Noodles	Las Acacias	0.5 Kg	0.43	Alimentos Las Acacias	2007/07
Peach jam	Dulciora	0.5 Kg	n/i	Arcor	2007/04
Peach jam	El Hogar	0.5 Kg	n/i	Lifibel SA	2010/11
Peach jam	Los Nietitos	0.5 Kg	n/i	Los Nietitos	2007/04
Peas	Campero	0.3 Kg	0.09	Regional Sur	2010/11
Peas	Cololó	0.3 Kg	0.09	Distribuidora San José	2010/11
Peas	Nidemar	0.3 Kg	0.09	Nidera	2010/11
Rice	Aruba tipo Patna	1 Kg	0.38	Saman	2007/04
Rice	Blue Patna	1 Kg	0.38	Coopar	2007/04
Rice	Green Chef	1 Kg	0.38	Coopar	2007/04
Rice	Pony	1 Kg	0.38	Saman	2010/11
Rice	Vidarroz	1 Kg	0.38	Coopar	2008/05
Rice	Saman Blanco	1 Kg	0.38	Saman	2010/11
Crackers	Famosa	0.14 Kg	0.28	Mondelez	2007/04
Crackers	Maestro Cubano	0.12 Kg	0.28	Bimbo	2007/04
Salt	Sek	0.5 Kg	0.09	Deambrosi	2007/04
Salt	Torre vieja	0.5 Kg	0.09	Torre vieja	2007/04
Salt	Urusal	0.5 Kg	0.09	UruSal	2007/04
Semolina pasta	Adria	0.5 Kg	0.43	La Nueva Cerro	2007/07
Semolina pasta	Las Acacias	0.5 Kg	0.43	Alimentos Las Acacias	2007/07
Semolina pasta	Puritas	0.5 Kg	0.43	Molino Puritas	2010/11
Soybean oil	Condesa	0.9 L	0.11	Cousa	2008/05
Soybean oil	Río de la Plata	0.9 L	0.11	Soldo	2010/11
Soybean oil	Salad	0.9 L	0.11	Nidera	2010/11
Sugar	Azucarlito	1 Kg	0.35	Azucarlito	2007/04
Sugar	Bella Union	1 Kg	0.35	Bella Unión	2007/04
Sunflower oil	Optimo	0.9 L	0.37	Cousa	2007/04
Sunflower oil	Uruguay	0.9 L	0.37	Cousa	2007/04
Sunflower oil	Río de la Plata	0.9 L	0.37	Soldo	2010/11
Tea	Hornimans	Box (10 units)	0.08	José Aldao	2007/04
Tea	La Virginia	Box (10 units)	0.08	La Virginia	2007/04
Tea	President	Box (10 units)	0.08	Carrau	2010/11
Tomato paste	Conaprole	1 L	0.16	Conaprole	2007/04
Tomato paste	De Ley	1 L	0.16	Deambrosi	2007/04
Tomato paste	Gourmet	1 L	0.16	Deambrosi	2010/11
Yerba	Canarias	1 Kg	0.64	Canarias	2007/04

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Table D.1 continued

Product	Brand	Specification	CPI share (%)	Owner / merger	Sample start (merger)
Yerba	Del Cebador	1 Kg	0.64	Molino Puritas	2007/06
Yerba	Baldo	1 Kg	0.64	Canarias	2010/11
Yogurt	Conaprole	0.5 Kg	0.13	Conaprole	2010/11
Yogurt	Parmalat (Skim)	0.5 Kg	0.13	Parmalat	2010/11
Yogurt	Calcar (Skim)	0.5 Kg	0.13	Calcar	2010/11
Bleach	Agua Jane	1 L	0.16	Electroquímica	2007/04
Bleach	Sello Rojo	1 L	0.16	Electroquímica	2007/04
Bleach	Solucion Cristal	1 L	0.16	Vessena SA	2007/04
Dishwashing detergent	Deterjane	1.25 L	0.13	Clorox Company	2007/04
Dishwashing detergent	Hurra Nevex Limon	1.25 L	0.13	Unilever	2007/04
Dishwashing detergent	Protergente	1.25 L	0.13	Electroquímica	2010/11
Laundry soap	Drive	0.8 Kg	0.45	Unilever	2007/04
Laundry soap	Nevex	0.8 Kg	0.45	Unilever	2007/04
Laundry soap	Skip, Paquete azul	0.8 Kg	0.45	Unilever	2007/04
Laundry soap, in bar	Bull Dog	0.3 Kg (1 unit)	n/i	Unilever	2007/04
Laundry soap, in bar	Nevex	0.2 Kg (1 unit)	n/i	Unilever	2007/04
Laundry soap, in bar	Primor	0.2 Kg (1 unit)	n/i	Soldo	2010/11
Shampoo	Fructis	0.35 L	0.36	Garnier	2007/04
Shampoo	Sedal	0.35 L	0.36	Unilever	2007/04
Shampoo	Suave	0.93 L	0.36	Unilever	2007/04
Soap	Astral	0.125 Kg	0.16	Colgate	2010/11
Soap	Palmolive	0.125 Kg	0.16	Colgate	2007/04
Soap	Rexona	0.125 Kg	0.16	Unilever	2012/12
Toilet paper	Higienol Export	4 units (25 M each)	0.24	Ipusa	2007/04
Toilet paper	Elite	4 units (25 M each)	0.24	Ipusa	2010/11
Toilet paper	Sin Fin	4 units (25 M each)	0.24	Ipusa	2007/04
Toothpaste	Pico Jenner	0.09 Kg	0.19	Abarly / Colgate	2010/11 (2012/07)
Toothpaste	Colgate Herbal	0.09 Kg	0.19	Colgate	2010/11
Toothpaste	Kolynos	0.09 Kg	0.19	Colgate	2010/11

Notes: Kg = kilograms; L = liters; M = meters; n/i = no information. Dates are reported as year/month; dates in parentheses are merger dates, following the original listing.